

University of South-Eastern Norway

Project Report

VIS3000 – Visualisering av Data

VÅR. 2025

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Visualisering av Data

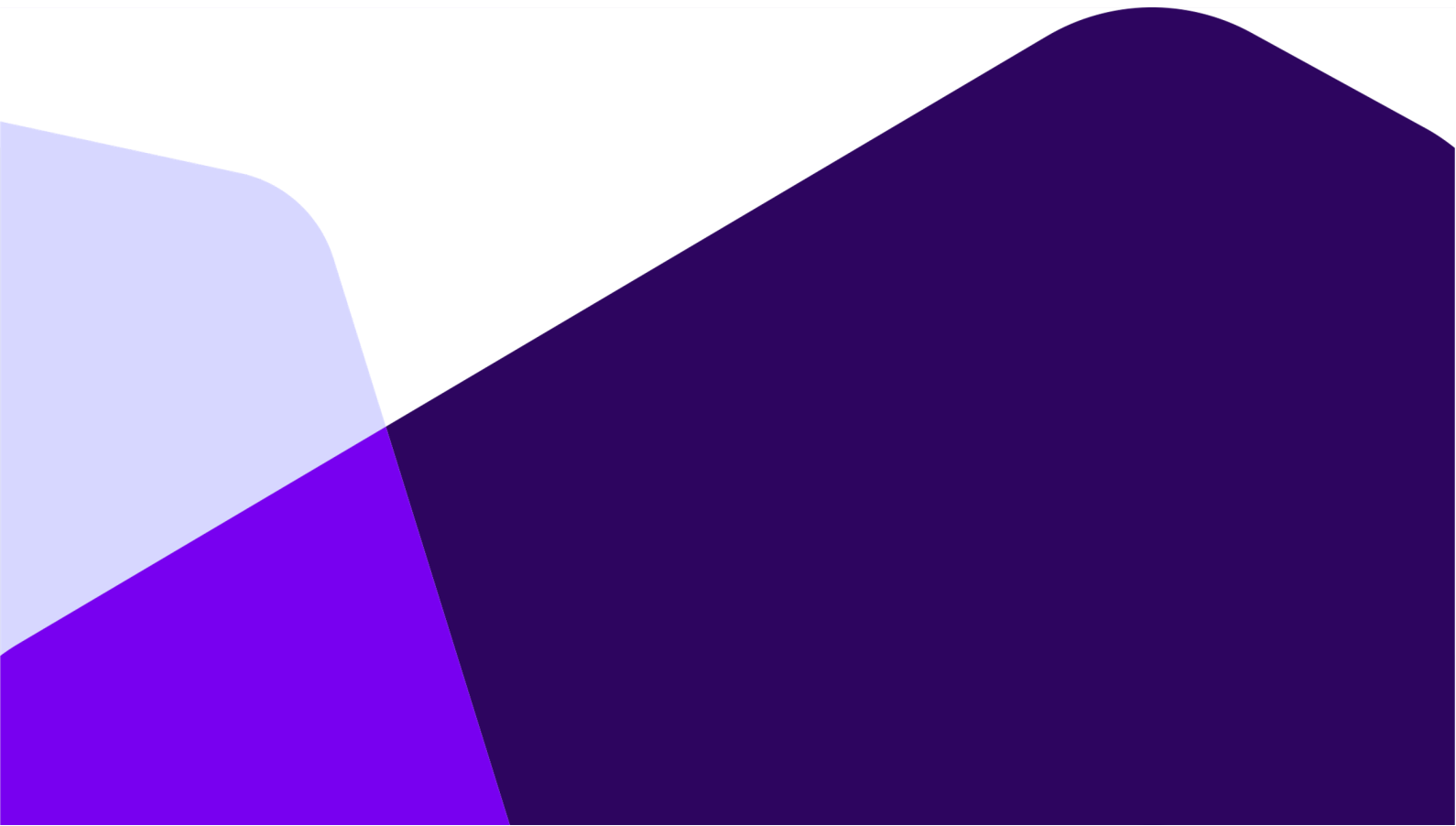


Table of Contents

1 Introduction	3
Background of the Topic	3
Problem statement	4
Research Questions:	5
Rationale of the problem statement	5
Target audience for this problem	6
Project goals	7
2 Dataset and preparation	7
Description of datasets	7
Data sources	10
Data cleaning	10
3 Methodology and Tools	10
4 Design Decisions for Visualizations	12
Design principles	12
Design choices - Colours and charts	13
5 Analysis and Findings	15
5.1 Time series analysis	15
Aging Population and the Need for Immigration in Norway	16
Prediction of Emigration and Net Immigration	17
5.2 Correlation analysis	21
5.3 How does this answer the research question?	21
6 Data Storytelling	23
Telling a story with key findings	23
Employment	30
Entrepreneurship	32
7 Dashboard	40
Description and purpose	40
Usability	41
8 Conclusion	42
9 Project management	43
7006	45
7047	46
7001 (Marziyeh)	46
7057	46
7054	47
10 References	47

Figure Table:

Figure 1: First part of infographic	p. 3
Figure 2: Second part of infographic	p. 4
Figure 3: Population Development and Age group Prediction in Norway	p. 18
Figure 4: Immigration, Emigration and Net Immigration	p. 20
Figure 5: Comparison of Immigration, Emigration and Net Immigration	p. 21
Figure 6: RStudio Correlation Men vs Women by Education level (Employed Only)	p. 23
Figure 7: Line chart, Life expectancy at birth of world	p. 25
Figure 8: World map visualization Life expectancy at birth of the world	p. 26
Figure 9: Population development and age group prediction	p. 27
Figure 10: Employment Rate by age and gender	p. 28
Figure 11: Number of users of care services	p. 29
Figure 12: Projection of the number of births	p. 30
Figure 13 Education level per year, excluding 62+	p. 31
Figure 14: Education level pr year for 62+	p. 32
Figure 15: R visualization faceted line plot: Education level and employment status	p. 33
Figure 16: RStudio Correlation: Men vs Women Education Level (Employed Only)	p. 33
Figure 17: R visualization bar chart: Education level vs Employment Status	p. 34
Figure 18: R visualization - Heatmap Amount of organizations started by immigrants	p. 35
Figure 19: Projected Immigration, Emigration and Net Immigration	p. 36
Figure 20: Total number of immigration by visa type and continent (Stacked Bar)	p. 37
Figure 21: Total number of immigration by visa type and continent (Line Graph)	p. 38
Figure 22: Total number of immigration by visa type and continent (Line Graph, Filtered by Asia)	p. 38
Figure 23: Total number of international students	p. 39

Figure 24: Employed in 2023 for third country students	p. 40
Figure 25: Norwegian View on Immigration	p. 41
Figure 26: Refugee Policy	p. 42
Figure 27: Dashboard	p. 44
Figure 28: Project team structure	p .47

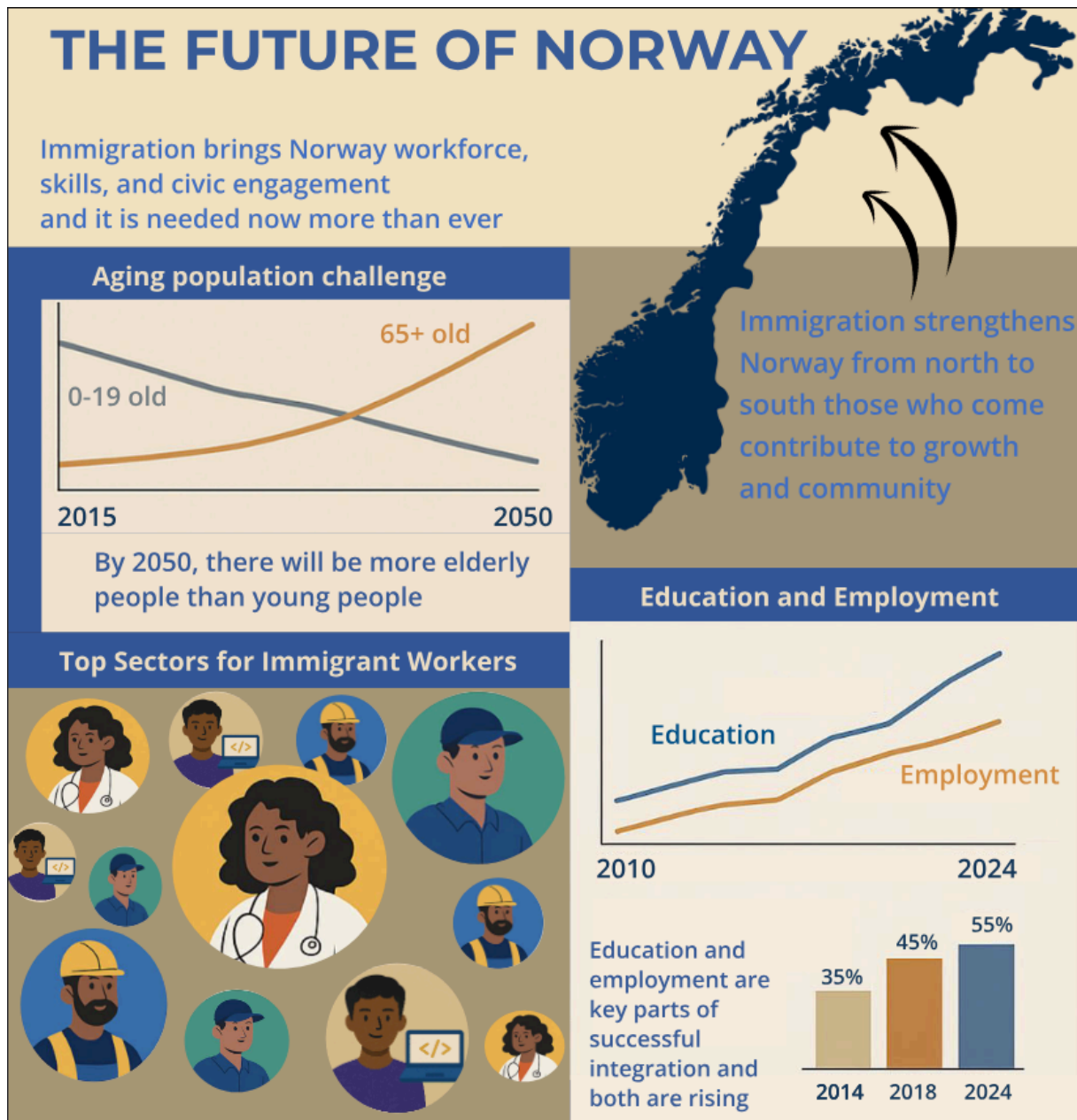


Figure 1: First part of infographic

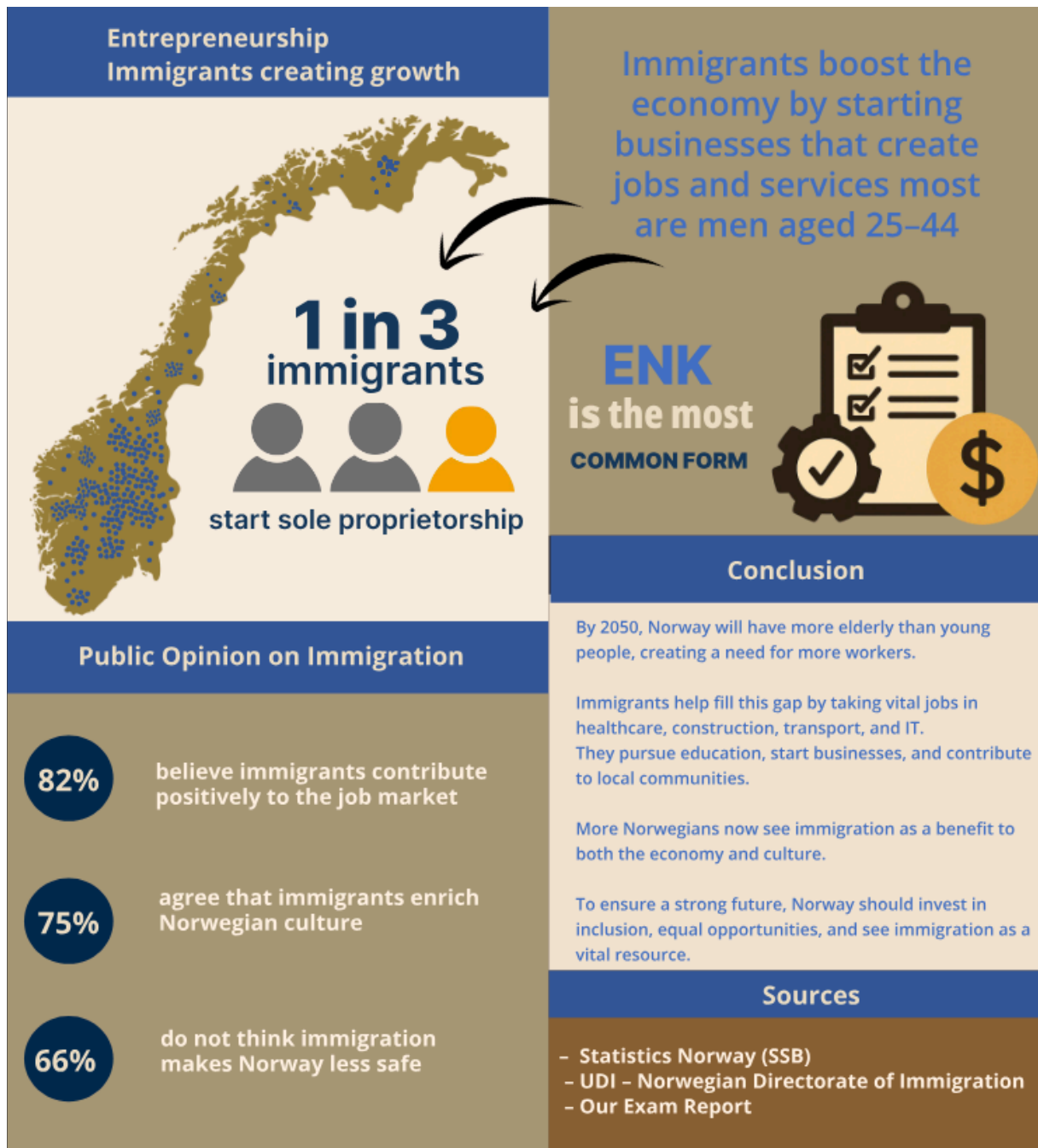


Figure 2: Second part of infographic

1 Introduction

Background of the Topic

Immigration has become a central topic in discussions about economic growth, social integration and demographic stability. Around the world, countries experience shifts in migration patterns due to economic opportunities, geopolitical conflicts and social factors such as aging populations and labor shortages. Norway is no exception. Over the past decades, immigration has played a crucial role in shaping the country's workforce, welfare system, and culture landscape. As Norway faces an aging population and growing need for skilled and unskilled labor, immigration is increasingly seen as a necessity for long term economic sustainability. This report will explore the primary reasons for immigration to Norway and analyze how these drivers have evolved over time. By examining how immigrants participate in the labor market, education, and society in general, the report will provide a structured understanding of how immigration has contributed to Norwegian society.

Understanding why people move to Norway is important for politicians, economists and society as whole. Immigration impacts many parts of daily life, from the job market and education system to healthcare and social services. Over the years Norway has seen changes in immigration patterns due to economic opportunities, global crises and political decisions such as joining the European Economic Area EEA (EØS), which allows free movement of workers from EU countries.

By examining immigration trends, we can gain a better understanding of how these patterns have changed over time and make predictions about how they may develop in the future, even though such forecasts are always uncertain. This knowledge helps shape future policies, improve integration efforts, and ensure that both immigrants and Norwegian society benefit from migration. Immigration is not just about numbers; it is about people, opportunities, and the future of Norway.

Problem statement

Life expectancy has gradually increased across the world, leading to a growing proportion of elderly people in almost all countries. Norway is among the countries with the highest life expectancy and is therefore facing a demographic shift where the elderly population will gradually make up a larger share than the younger generations. This means that the number of people of working age will decline relative to the number of elderly.

At the same time, we know that elderly people have greater needs for health services. An increase in the number of elderly will therefore place additional pressure on health and care services. This development creates a double challenge: on the one hand, there will be fewer people in the workforce, and on the other hand, the need for labor—especially in the health sector—will increase.

To address this challenge, Norway must think and plan strategically to reduce the gap that may arise in the labor market in the years ahead. Several measures can be considered. One option is to increase labor participation, for example by raising the retirement age or increasing the number of hours worked per person. Another option could be to increase birth rates so that more young people eventually enter the workforce.

However, both of these solutions are demanding. Raising the retirement age is politically difficult, and increasing birth rates is a long-term solution with uncertain outcomes, especially since current statistics show that birth rates continue to decline.

A third option is to increase immigration. This appears to be the most realistic solution in the short term, but it also comes with challenges. Figures from Statistics Norway show that although immigration has increased, net immigration is expected to decline because more people are also choosing to emigrate.

Norway therefore faces a complex challenge that cannot be solved easily or quickly. This highlights the need for long-term and comprehensive planning to ensure that society is better prepared when the aging population begins to have significant consequences for both the labor market and welfare services.

Research Questions:

We aim to explore this issue further and examine the immigration solution in more detail. To support this process, we have developed a set of questions that will serve as a guide throughout the project:

1. How has Norway's population growth changed over the past decades, and what are the projected demographic shifts according to Statistics Norway (SSB)?
2. How do immigrants contribute to society? Are there groups that integrate more quickly and enter the workforce sooner? (How do different immigrant groups in Norway vary in terms of workforce participation within the first five years of arrival?)
3. How are immigration, emigration, and net immigration expected to develop in Norway over the next 25 years, and what potential implications might these trends have for population growth and the labor market?
4. What immigration and emigration trends can be observed in Norway over the past 10 years, particularly in relation to region of origin and visa type?
5. What are Norwegian citizens' attitudes toward immigrants and immigration, and how have political debates influenced these views over time?

Rationale of the problem statement

The reason we chose this problem is because it is something that will affect everyone living in Norway in the years ahead. As the population gets older and there are a lot less young people entering the workforce, we face serious challenges, especially in areas like healthcare, elderly care and public services (NordForsk, 2022).

We wanted to understand what kind of options Norway actually has to deal with this situation. The three strategies we looked at aimed at increasing the overall labor effort, increasing the birth rate and using immigration (AARP International, 2024). These are all things that are often talked about in politics and the media. But it is not always clear how realistic they are or how much they can actually help in the long run.

We thought it was especially interesting that immigration is often mentioned as a solution, but at the same time the data from SSB shows that net immigration is going down. That did not really match up, so we wanted to look more closely at what is happening. We also saw that there was no clear data on how emigration might develop in the future, so we made our own prediction and used it to see what that could mean.

By showing these trends with visualizations and putting the numbers into context, we wanted to make the topic easier to understand. Not just for ourselves, but for people who work with this in real life, like politicians, NAV or people working in integration and healthcare.

Target audience for this problem

The people we had in mind when working on this project are the ones who deal with population change, work and immigration in practice. That includes people working at NAV, in local government, in healthcare services and in planning. It also includes politicians and others who help shape future policy.

These people need to understand how immigration actually works in the long run, not just in numbers, but how it affects things like the labour market, public services and population growth (Ministry of Labour and Social Inclusion, 2023). We believe that good visualizations and clear data can help them make better decisions.

We also think that this topic is important for anyone who wants to understand how Norway will change in the coming years. Immigration is not just a political issue, it is something that will affect schools, workplaces, the economy and the way people live together in society.

Project goals

The main goal of this project is to take a large dataset and turn it into something that is easy to understand by using clear and useful visualizations. We use tools like Tableau and RStudio to create dashboards, infographics and interactive visuals that explain complex data in a simple way.

The specific goals of the project are:

- To collect, prepare and analyze data from reliable sources
- To explore patterns, trends and connections in the data
- To use good visualization techniques that support storytelling and help people make better decisions
- To design dashboards and graphics that are easy to use, informative and nice to look at
- To share insights that help others understand the topic better

In short, the project is about combining data and design to tell a story that matters.

2 Dataset and preparation

Description of datasets

To get an overview of what we are working with, with the datasets, we have made this table to show each variable, its description and which scale the variable used. This is done to have an early understanding of the data we are going to work with and how it is used.

Variable	Description	Scale used
Year	The years the data is related to	Interval scale (2013-2024, 2024-2050, 1770-2023)
Amount	Number of people	Ratio Scale (0 and up)
Gender	Genders of the individuals in the data	Nominal Scale (Men, women)
Age	Age range of the individuals	Ordinal Scale (Ordered categories, 15-29 etc.)
Education Level	The education level of the individuals in the data	Ordinal Scale (Elementary, High School, etc)
Status	Employment Status	Nominal Scale (employed only, unemployed etc)
Organizationform	The type of organization form	Nominal Scale (ENK, ANS etc.)
View on Immigration	Public opinion on immigration's impact on society (3 subcategories: Cultural Enrichment, Safety Concern, Workforce	Ordinal Scale (Strongly Agree, Somewhat Agree, etc.)

	Contribution)	
Refugee Policy Opinion	Public opinion on whether refugee policy should change	Ordinal Scale (Easier, Same as Today, Harder)
Continent	Different continents of the world	Nominal Scale (Africa, Asia, Europe,...)
Country	Different countries of the world	Nominal Scale (Albania, Algeria, ...)
Total number of students	Number of people	Ratio Scale (0 and up)
Number of birth	Projection of number of birth (2023-2050)	Ratio Scale (0 and up)
Visa Type	Visa Type for immigration to Norway	Nominal Scale (Family, Work, ...)
Number of users in percent (per 1000 inhabitants)	Number of users of care services (per 1000 inhabitants)	Ratio Scale (0 and up) in percent type
Predicted Immigration	Forecasted emigration based on historical trend	Ratio Scale (0 and up)
Net Immigration	Immigration minus Emigration per year	Ratio Scale (can be 0 or negative)
Emigration	Number of people leaving Norway per year	Ratio Scale (0 and up)
Employment Rate	Percentage of people employed	Ratio Scale (0 and up)

Our study was based on a number of government documents that offer detailed information about immigrants, employment, education, company ownership, and demographic shifts in the world and Norway. The Norwegian Directorate of Immigration (UDI), Statistics Norway (SSB) and Our World in Data websites provided the datasets, which were downloaded and

made available in structured formats, such as CSV and Excel files. Each row represents a statistical observation related to a specific year, and the columns include characteristics such as gender, age group, country of origin, type of residence permit, work status, educational attainment, and the number of participants.

While formats varied slightly depending on the thematic focus, most datasets included time series data spanning several years, allowing us to follow the evolution of various aspects of immigration over time. While some offered national data in aggregate form, others were broken down by age and gender. When needed, we broadened the initial classifications into more inclusive groups to ensure consistency across the datasets. This was important for combining or comparing data on several topics, including demographics and immigration motivation or education and employment.

Data sources

We have specifically chosen these as our primary source because of their official status, extensive coverage, and proven reliability. Both of these institutions are managed by the Norwegian government, which ensures that the data they collect is of a high standard, verified and transparent. Since they are directly involved in immigration and demographic statistics, we are confident that they are able to provide the most accurate data for our project.

Data cleaning

We had to transform the data to fit our analysis and visualization tools using Tableau Prep. This involved restructuring the datasets by using the pivot tool to convert between wide and long formats, depending on what was most suitable for the analysis. We also created new columns by applying operations such as aggregate, join, and union. Examples include generating a continent column based on country, and combining several protection categories under a single "protected visa type" column. Additionally, we changed column names from

Norwegian to English when needed, for example, when we wanted to compare multiple variables or highlight temporal developments.

We also built custom datasets by combining selected columns from multiple tables, particularly when working with immigration numbers by visa type and country. This proved to be a time-consuming task, as we often had to try different combinations before finding dataset structures that actually supported our problem statement.

For specific analyses, such as exploring the relationship between education and employment, we merged datasets based on common identifiers like year and gender. This data preparation phase laid the foundation for our analysis, allowing us to uncover complex relationships across multiple domains. At the same time, it ensured high data quality and enabled us to present our findings through clear, reliable, and visually appealing visualizations in Tableau Desktop.

3 Methodology and Tools

In this project, we mainly used Tableau, Tableau Prep, and Excel to explore, clean and visualize our data. These tools were chosen not only because they are powerful and user friendly, but also because they allowed us to transform complex information into clear insights that could be easily understood by a wide audience.

We started with Excel to get the first impression of the datasets we had collected. Since most of the raw data came from SSB and UDI, it was delivered in CSV and Excel format. Using Excel helped us quickly review the structure, identify missing values, remove irrelevant columns, and do some basic filtering and sorting. It was a simple but effective way to organize the data before moving on to more advanced tools.

Once we had cleaned up the data, we moved into Tableau prep for the more technical part of data preparation. Our datasets came from different sources and covered a range of topics such as education, employment, visa types etc. To make them work together, we

had to translate column names from Norwegian to English, adjust formats, and merge datasets using common variables like year, gender and age group. Tableau prep made this process intuitive and visual, which helped us keep track of the structure and spot inconsistencies early on.

The core of our analysis and storytelling was done in Tableau. This is where we built most of the visualizations included in our report, line charts to show changes over time, bar charts to compare opinions or visa types, and area charts to distribution of immigration categories. Tableau allowed us to build an interactive dashboard that combined multiple visual elements in one place. This helped us highlight key trends, reveal interesting correlations, and make data engaging for the viewer. We also used Tableau's features like trend lines and forecasting to illustrate demographic shifts, such as the growing elderly population in Norway and the potential labor shortages that may follow.

One of the things we appreciated most about Tableau was how it supported storytelling. The tool let us design visualizations that not only presented data, but also told a story about why people come to Norway, how they integrate, and what that means for the country's future. This was especially important because one of our goals was to inform decision makers, organizations and the general public about the realities of immigration in a balanced and evidence based way.

Alongside these tools we also made use of R and RStudio to help with analyzing trends using correlation methods and creating some visualizations. We primarily used the tidyverse packages like dplyr and tidyr to help with computing correlations, which we will showcase later on. To create visualizations we made use of the ggplot2 package to produce trend charts, bar charts and heatmaps to visualize the correlations alongside simple visualizations of the data.

Infographic was designed using Figma, based on data analysis and visual storytelling techniques. Key findings were selected, structured, and visualized to communicate a clear message supported by both statistics and narrative flow.

All in all, using these tools gave us a solid workflow from raw data to finished story. Excel helped us get started, Tableau Prep helped us clean and organize, and Tableau and RStudio helped us analyse and communicate. Together they made it possible to turn large, sometimes messy datasets into meaningful insights that are easy to explore and understand.

4 Design Decisions for Visualizations

Since our dataset is mostly made up of time series data, we had to think of the best way to make sure that we can show the trends and progressions over time in the best way possible, for readability and simplicity.

Design principles

To achieve our goals with visualization and our storytelling we have had to implement some design principles.

Clarity and Simplicity:

We want to have a clear and simple design which allows for easy reading and understanding, helping us reach our goals in providing insights into immigration in Norway. With this principle in place we can help our readers in understanding the essential details of our data.

Consistency:

Being consistent across several visualizations helps us in keeping the story we want to tell in being coherent and as concise as possible. This will give us a greater impact and prevent confusion in the minds of the reader.

Accessibility:

Using colours in our visualizations with contrasting hues and transparency effects, we allow

our designs to also reach those that are affected by colour blindness and makes the results be accessible to a wider range of people.

Data representation:

Selecting the best types of charts for our data, line and area for trends over time, bar charts for comparisons helps us ensure that our visualization aligns with the dataset that we have.

Design choices - Colours and charts

Line graphs and line plots:

This type is chosen because they are a good visualization to show the progression over time for the data. It is also easy to read and intuitive for most readers., showing the trends quickly and efficiently. To differentiate the lines we have chosen to implement different colours. This is to make sure that the reader understands the difference between what the lines represent.

The different lines are used to display the genders or age groups. This colorization is also displayed in the diagrams to show the reader what represents what.

The colors that have been chosen had to be contrasting to prevent issues with similarity, which may cause confusion for the reader. The orange and blue colors have been used when there are two lines displayed. When increased lines are displayed we added red and green lines as well.

We encountered some issues in displaying all the data in a single visualization, especially in the education, employment and entrepreneurship sections. This was caused by the major difference in data amounts between the different age groups or categories. To solve this issue we decided to separate the different age groups where it made sense. This way we were able to show the progression for each age group within the different categories.

Bar chart:

Complementing these line diagrams are the bar charts, which displays the amount of data for each category. This is used to display the difference in opinion of the Norwegian population on topics regarding immigration and immigration policies. This visualization is best used when the reader is supposed to compare different categories, as this shows the majority for

each category in a simple and easy way. It is also able to show a progression or trend over time, if there is an increase or decrease in the data amount for each category.

Using colors here to differentiate the different categories, helps the reader to understand which category has the largest amount and what it represents.

Area graphs:

We have also chosen to use area graphs to display the different types of visa for immigration to Norway in a time period. This layered area chart is chosen to show the reader the amount for each of the visa types and compare them with each other in a simple and clean way. Using the layered effect allows us to show the major differences in a single view, overlapping the different categories rather than summing them up. This choice helps us to view each visa type separately while still allowing us to show us the direct comparisons between them.

The colors are relatively similar but are contrasting enough to show the differences, using the transparency effects on the layers helps with the contrasting. With the yellow, orange and red colors it is not too contrasting so that it is a strain for the readers.

Heat map:

We chose to make use of the heat map for the section about entrepreneurship as this is a simple and clear visualization for readers to understand. This is especially useful when we try to compare different amounts of something between different categories, like age, organization form or genders. We chose to use colder colours to display this as they are not so intrusive for the eyes and it is easy to differentiate between them as the darker the colour the larger amount.

Other chart types:

We have also looked at other different chart types, but we thought they would not be adequate to achieve our goals. One of these were the stacked bar charts. This chart type became too cluttered when seeing all the different categories, genders and years together, in the employment section. We also tried to use the area chart for the education section, but because of the difference in data amounts in the different age groups it was not good enough to be displayed clearly.

5 Analysis and Findings

5.1 Time series analysis

Using data from Statistics Norway (SSB) and visualization made in Tableau, we can see how the population in two age groups, 0-19 years and 65+ years is changing from 2013-2050. We imported the dataset into Tableau and created a line chart that shows the number of people in two age groups. Then we used linear regression to estimate how the population in both groups could develop until 2050.

In Tableau, we applied a linear trend line to both groups and extracted the equations used for prediction:

- For the 0–19 age group, the regression formula was:

$$\underline{y = -2423.72 * x + 6,142,580}$$

- For the 65+ age group, the formula was:

$$\underline{y = 22070.1 * x - 43,638,300}$$

Here x represents the year, and y is the predicted number of people in that age group. These formulas allowed us to create calculated fields in Tableau, so we could extend the chart and visualize the expected population numbers 2024 to 2050. We also added a vertical line at 2023, which marks the end of the real data and the beginning of the prediction. The predicted values are shown with dashed lines to make them visually different from the actual data.

Aging Population and the Need for Immigration in Norway

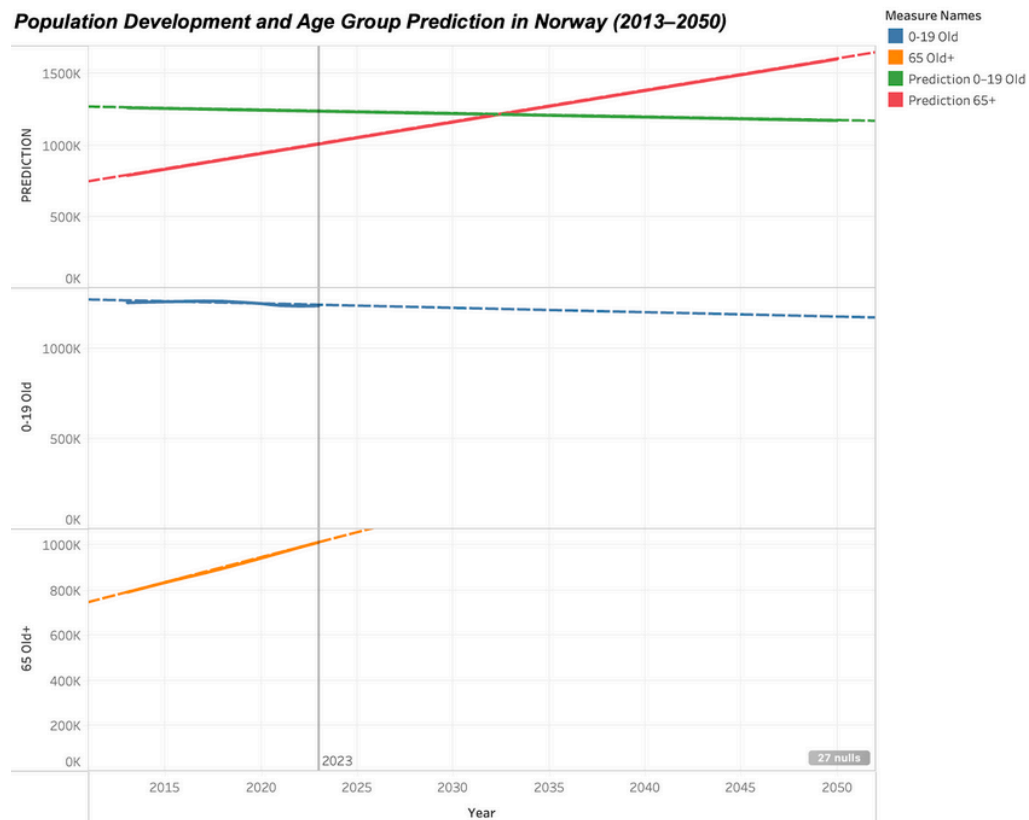


Figure 3: Population Development and Age group Prediction in Norway

The final visualization shows a clear trend:

The number of people aged 65+ will increase sharply toward 2050.

The number of young people stays stable or slightly decreases.

Around the year 2040, the two lines cross, meaning there will be more elderly than young people.

This gives a strong visual picture of the aging population in Norway and why this is a problem.

Prediction of Emigration and Net Immigration

To understand how Norway’s net immigration might develop toward 2050, we used both official data and our own predictions.

SSB provides population projections for immigration and net immigration, so we used those directly. However, they do not publish separate projections for emigration. Because of that, we created our own prediction based on historical data from 2010 to 2024. We assumed emigration will gradually increase, since more people are moving out of Norway for work, studies or to return to their home country.

By combining SSB’s immigration data with our emigration prediction, we were able to calculate a more realistic estimate of future net immigration.

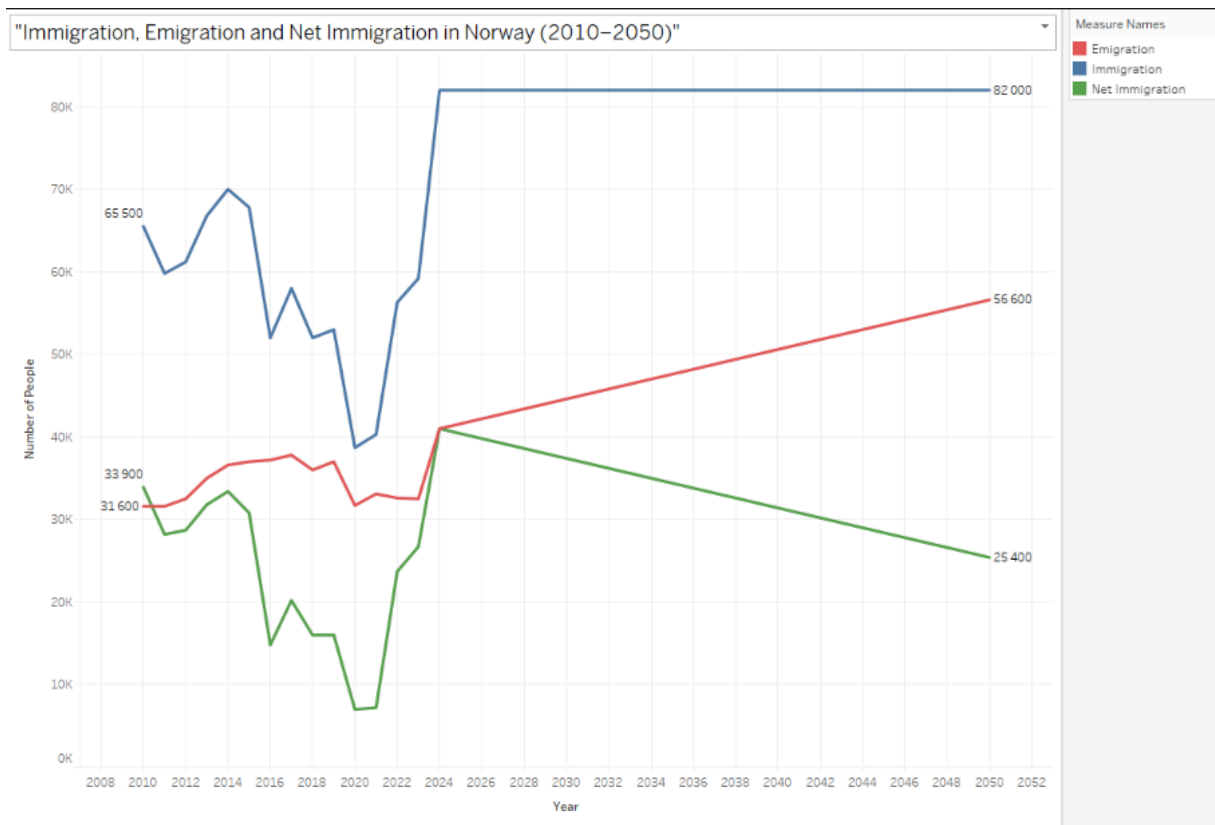


Figure 4: Immigration, Emigration and Net Immigration

Figure 4 shows that while immigration stays stable at around 82,000 people per year, emigration increases steadily – from around 30,000 in 2010 to over 55,000 by 2050. As a result, net immigration drops from more than 41,000 in 2024 to just around 13,000 in 2050.

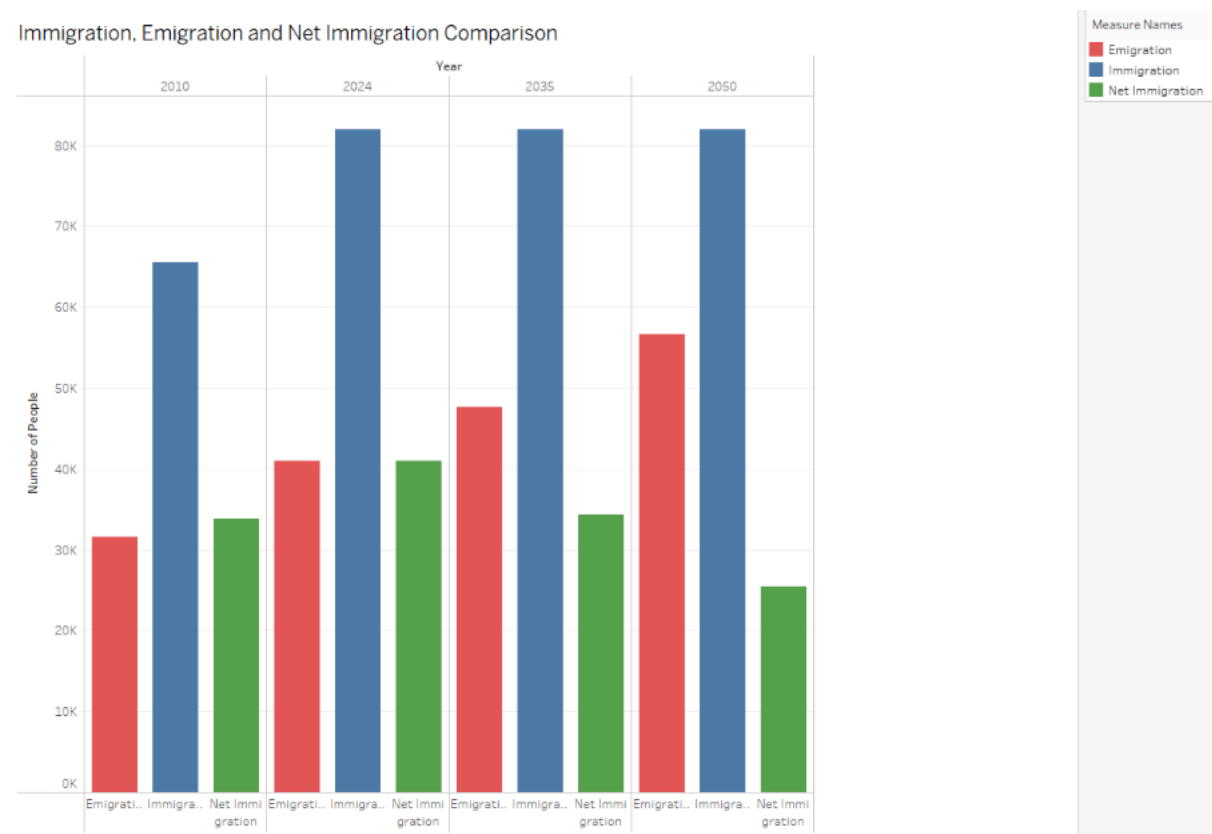


Figure 5: Comparison of Immigration, Emigration and Net Immigration

Figure 5 shows this change more clearly in key years: 2010, 2024, 2035 and 2050. The gap between people coming in and moving out is getting smaller. This means that even if immigration stays high, the population growth from immigration will slow down unless Norway also finds ways to retain immigrants over time.

Norwegians have grown more accepting of immigration in terms of employment and culture throughout time. Positive sentiments showed a modest decline between 2023 and 2024, although the overall pattern is evident. More individuals view immigration as a positive issue than a negative one. The rise in worries regarding refugee and safety measures in 2024, however, indicates that attitudes are still evolving (Bartsch, 2024).

Norway faces several challenges related to immigration and labor market demands. To ensure sustainable development, it is crucial to analyze which types of visas yield the most positive effects. This report compares three main visa categories, work, education, and family visas, based on available immigration data.

People move to Norway for different reasons, like work, school or to be with family. A big part of the immigration is related to jobs, especially from other European countries. Many come from Eastern Europe because the EEA agreement makes it easier for them to live and work here (OECD, 2022).

Education-related immigration is a smaller group, often with people coming from Asia or North America to study. Family immigration is more spread out across the world and often happens because someone already has family living in Norway (Statistics Norway, 2024).

People who come to Norway with a work visa usually get into the job market quickly and help contribute to the economy. Most of them also have a high chance of getting a job. But in some jobs with low pay or short contracts, it can be harder. That can make things a bit unstable, both for the workers and for the job market.

Education visas attract highly skilled individuals who can potentially stay in Norway after completing their studies (Bartsch, 2024), but there is a need for improved incentives to retain this talent, ensuring that skilled professionals contribute to the national workforce.

Family visas play a vital role in integration and social stability but generate less immediate economic benefit compared to work visas. If individuals arriving under this category do not enter the labor force, there can be increased pressure on welfare services (Ministry of Labour and Social Inclusion, 2023). Thus, policy measures should focus on ensuring higher employment rates within this group.

5.2 Correlation analysis

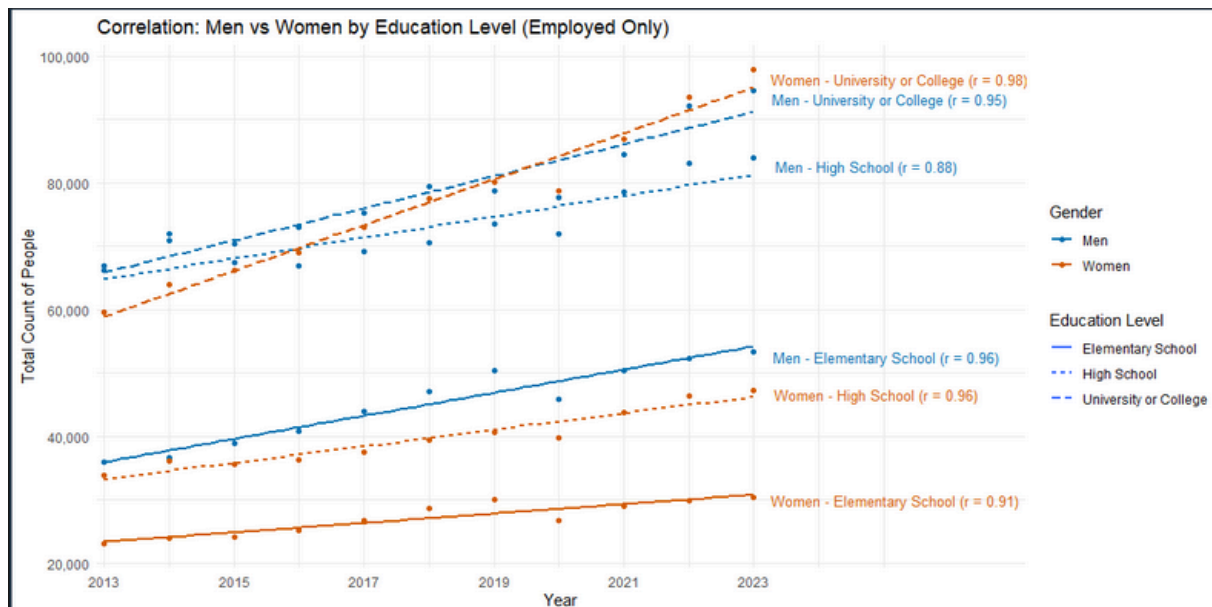


Figure 6: RStudio Correlation Men vs Women by Education level (Employed Only)

We have chosen to make use of Pearson Correlation method to find correlations in our data. The method takes two continuous variables and measures the strength and direction relationships between the two. This correlation, r , will see if there is any statistical evidence found in the data. (Kent State University Libraries, 2025) In the graph above we looked at the correlation between education levels and being employed. The results showcase a strong correlation between having a higher education leads to higher employment. There is also a strong positive trend across all levels which is great. We can also see that the gender gap at the top educated immigrants is narrowing with the women jumping ahead of the men, while the gap is very clear in the High School and Elementary School levels with women showing lower numbers of employment. Example is that immigrant men with an elementary school education have slightly more employment than immigrant women with a high school education, which is a worrying statistics even though it is trending upwards slightly.

5.3 How does this answer the research question?

RQ1 and RQ3 are covered in the time series analysis which shows that Norway's population is aging and that the age group 65+ will overtake the 0-19 age group by around 2040. This suggests that the idea of immigration is essential to create some stability to the labor market. Our projections also show a worrying trend, gross immigration is expected to remain stable but emigration is likely to rise, which leads to a drop in net immigration. This directly covers the RQ3 and showcases the importance of maintaining growth in the population and

workforce will require more than just gaining new immigrants but also to help them when they get here, so that they stay.

RQ2 which covers the workforce participation of immigrants is addressed in the correlation analysis which shows a strong positive relationship between education level and number of employment among immigrants. This is especially clear when looking at immigrant women with a higher education, with employment numbers equalling or even surpassing the immigrant men. This suggests that better and higher education and integration programs directly affect improved workforce participation, which answers the RQ2 about which groups integrate more quickly.

RQ4 is answered through the visa type analysis which reveal that labor immigration, from Europe especially, is the most significant contributor to the workforce. Immigrants with an educational visa show strong long-term potential, but there would be a need for policy changes and incentives for them to remain in the country after concluding their education. These trends support the need to tailor immigration policies based on what kind of visa type they have and regional origin.

RQ5 regarding the public's attitudes, the data show that Norwegians in general are becoming more accepting immigrants, while there are some worries about refugee policies and security in recent years. The findings show that the public's opinion will continue to affect immigration policies and the success rate of integration.

6 Data Storytelling

Telling a story with key findings

According to the World Health Organization (WHO), all countries around the world are experiencing a growing number of elderly people in their populations. In 2024, the number of people aged 60 and over increased to 1.4 billion, up from 1 billion in 2020. By 2050, this number is expected to double from the 2020 level, reaching 2.1 billion (World Health Organization, 2024).

One of the key drivers behind this demographic shift is the steady increase in life expectancy. The chart below shows how global average life expectancy has risen significantly from 1770 to 2023, reflecting improvements in healthcare, living conditions, and medical advancements.

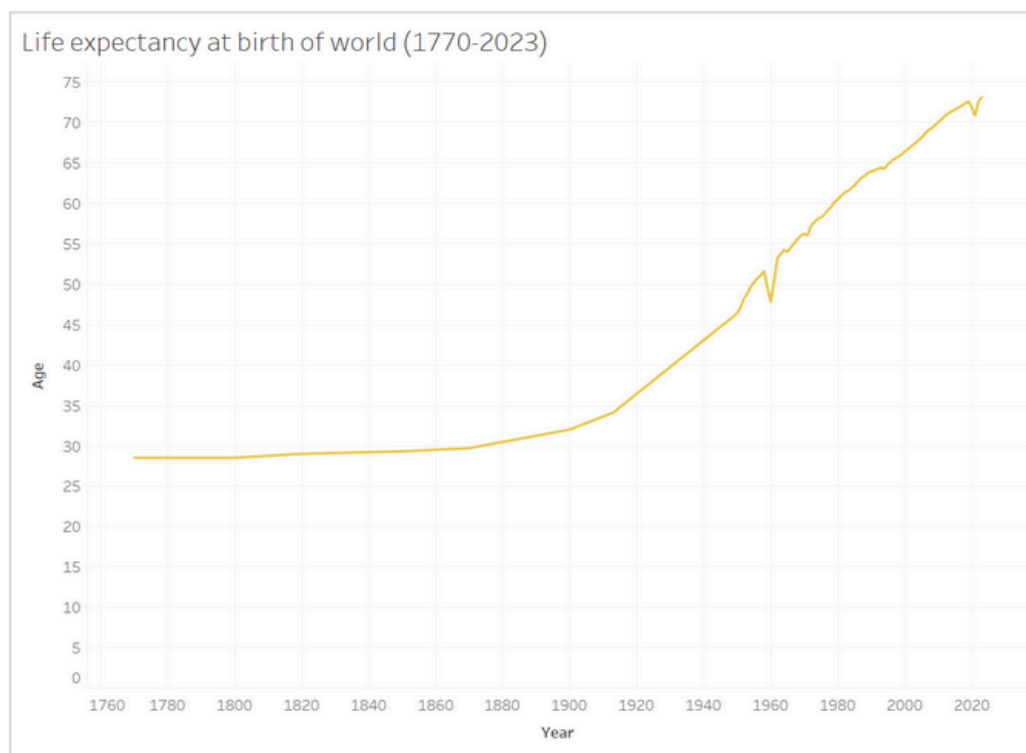


Figure 7: Line chart, Life expectancy at birth of world

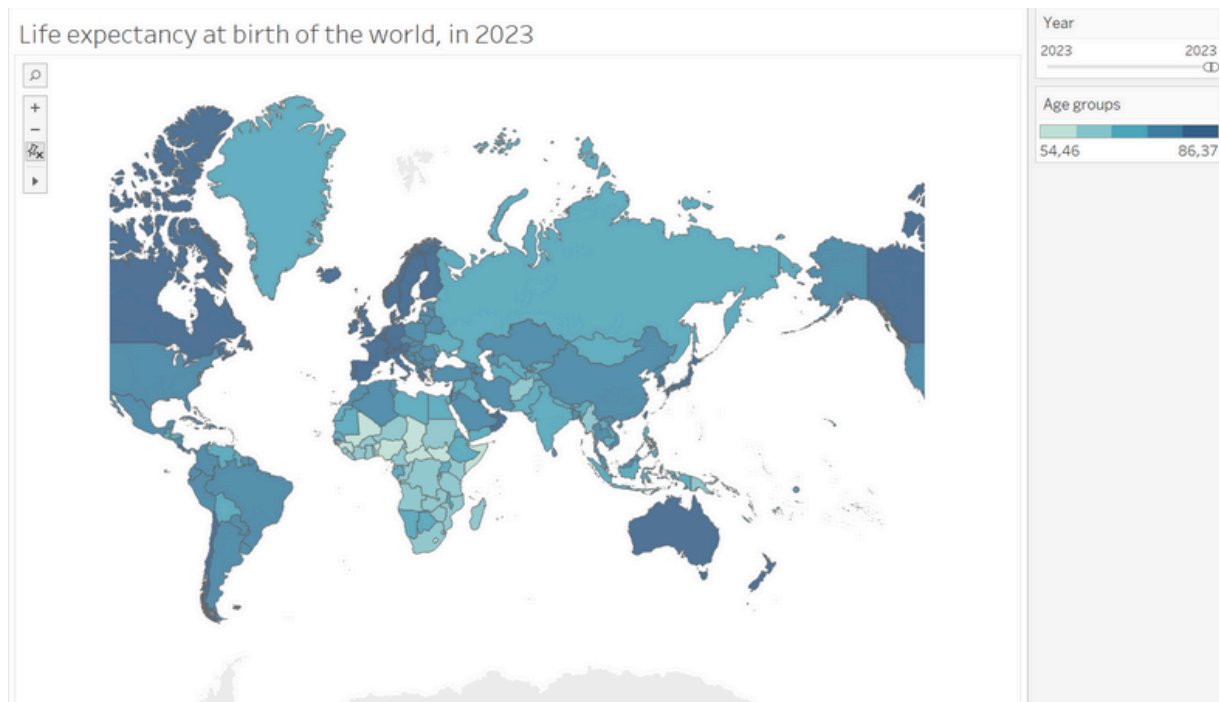


Figure 8: World map visualization Life expectancy at birth of the world

The heat map shows life expectancy at birth across countries around the world. According to the map, several countries in Africa have the lowest life expectancy, while many other countries in Europe, the Americas, and Australia are among those with the highest (Herre, 2025).

Norway is among the countries with the highest life expectancy in the world, as shown in the map. This reflects strong public health, high living standards, and access to quality healthcare.

However, high life expectancy also means that more people live into older age groups. Norway is not exempt from this demographic shift. Over the coming decades, the number of people aged 65 and over will exceed the number of children and young people (aged 0–19). This aging trend is both strong and clear, indicating that Norway’s population is growing, but also becoming significantly older.

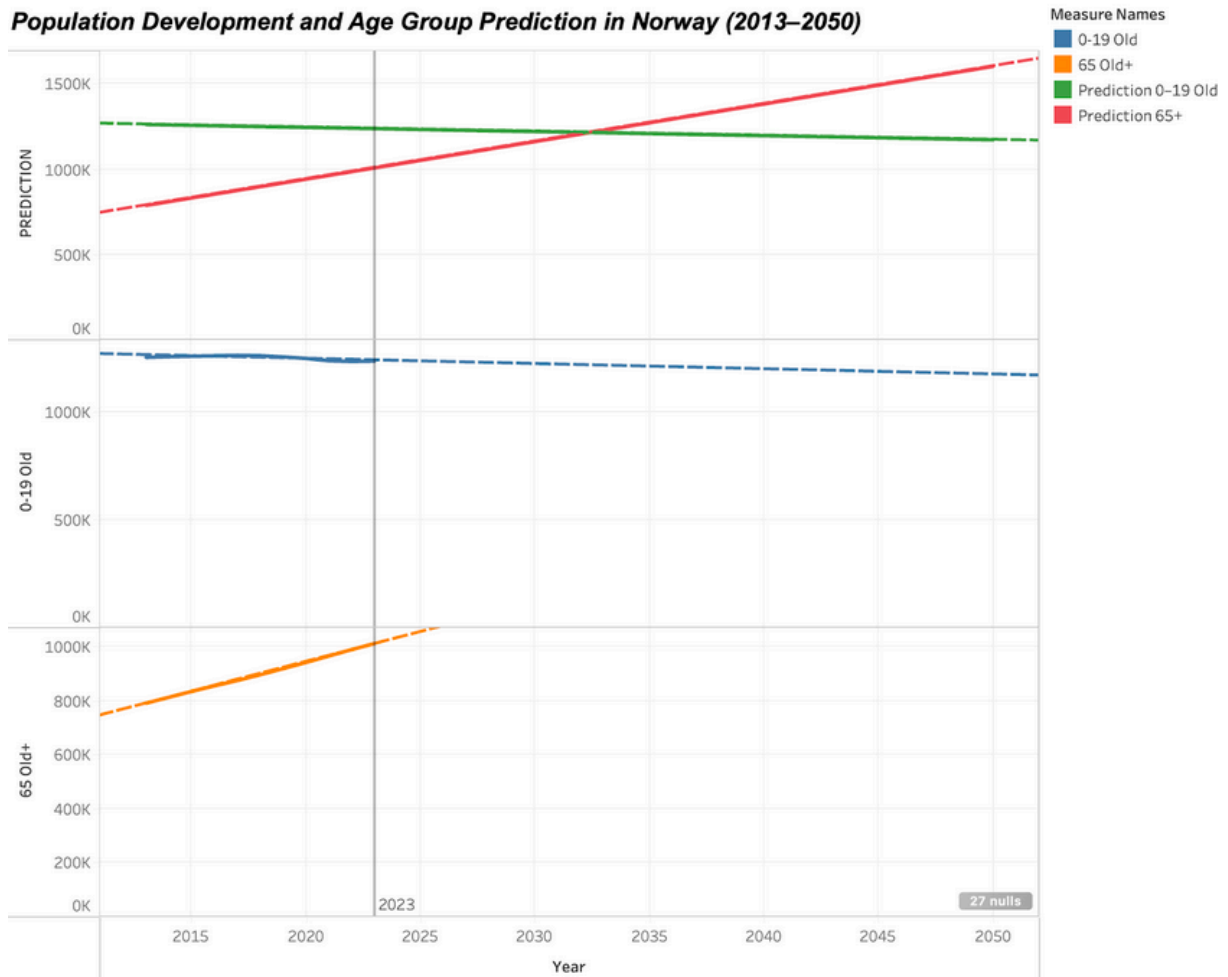


Figure 9: Population development and age group prediction

The line chart, based on data from Statistics Norway (SSB), visualizes the population development in the two age groups (0–19 years and 65+ years) and clearly illustrates the aging trend in the Norwegian population.

This aging trend has significant consequences for Norway, particularly for the labor market. Based on data from Statistics Norway (SSB), the employment rate is low among younger individuals (15–19 years), increases with age, and then stabilizes between the ages of 25 and 49. After age 50, it gradually begins to decline, and from age 65 onward, we observe a sharp drop in employment rates.

Employment Rate by Age and Gender in 2024

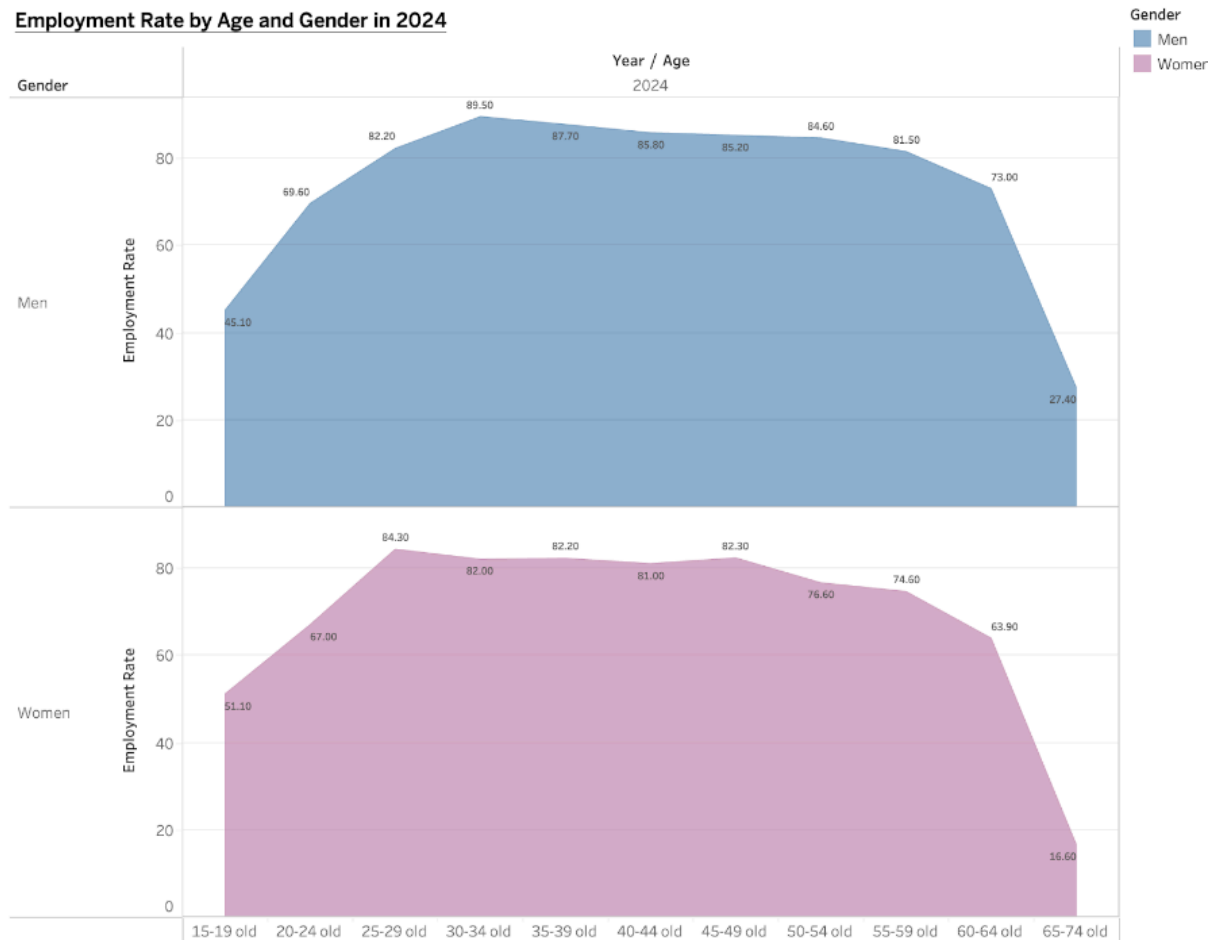


Figure 10: Employment Rate by age and gender

This chart highlights a key challenge: as more people move into the older age groups, where employment is significantly lower, we will have fewer people participating in the workforce.

In addition, data from Statistics Norway (SSB) shows that older age groups use significantly more healthcare services than younger ones.

Number Users of care services (per 1000 inhabitants)

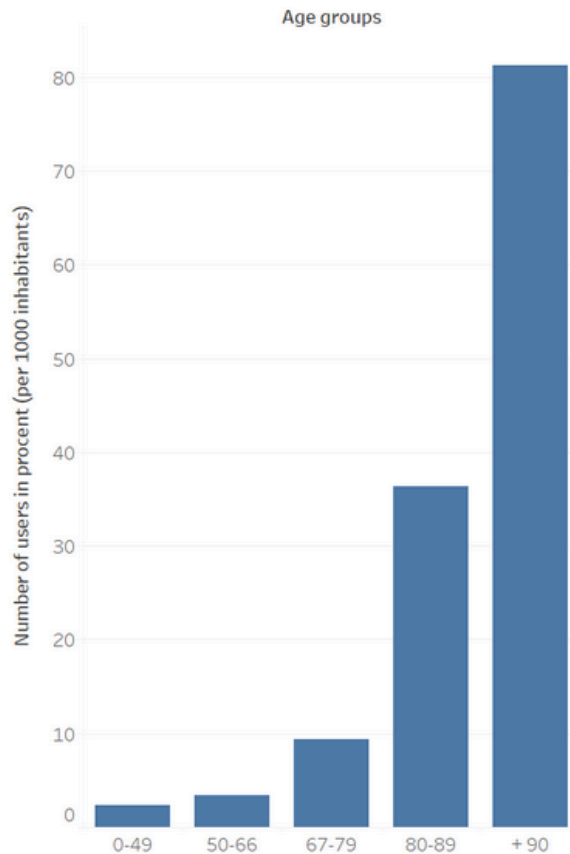


Figure 11: Number of users of care services

The Bar chart displays the percentage of healthcare service users per 1,000 inhabitants across different age groups. It clearly illustrates that the need for healthcare increases gradually after the age of 50, and rises sharply after age 80.

The combination of a shrinking working-age population and a growing demand for healthcare highlights a critical challenge: the need for more workers, especially in the health and care sectors, will increase significantly in the coming decades. This means that Norway must take long-term planning seriously to address these demographic changes. To respond to this challenging trend, three potential solutions have been discussed.

The first involves increasing labor participation, for example by reducing sick leave rates, lowering the number of people on disability benefits, raising the retirement age, or increasing working hours. However, these measures are politically sensitive, as reflected in the ongoing debates about absenteeism, the weakening of the workfare model, and adjustments to the retirement age in line with increased life expectancy.

The second is to raise the birth rate so that future generations can help stabilize the labor market. Yet current trends show a continued decline in fertility, and even if birth rates were to rise again, the positive effect on the workforce would not be felt for several decades. The Area chart below shows a projection of the number of births from 2023 to 2050, indicating a downward trend (Skogheim.K, 2024).

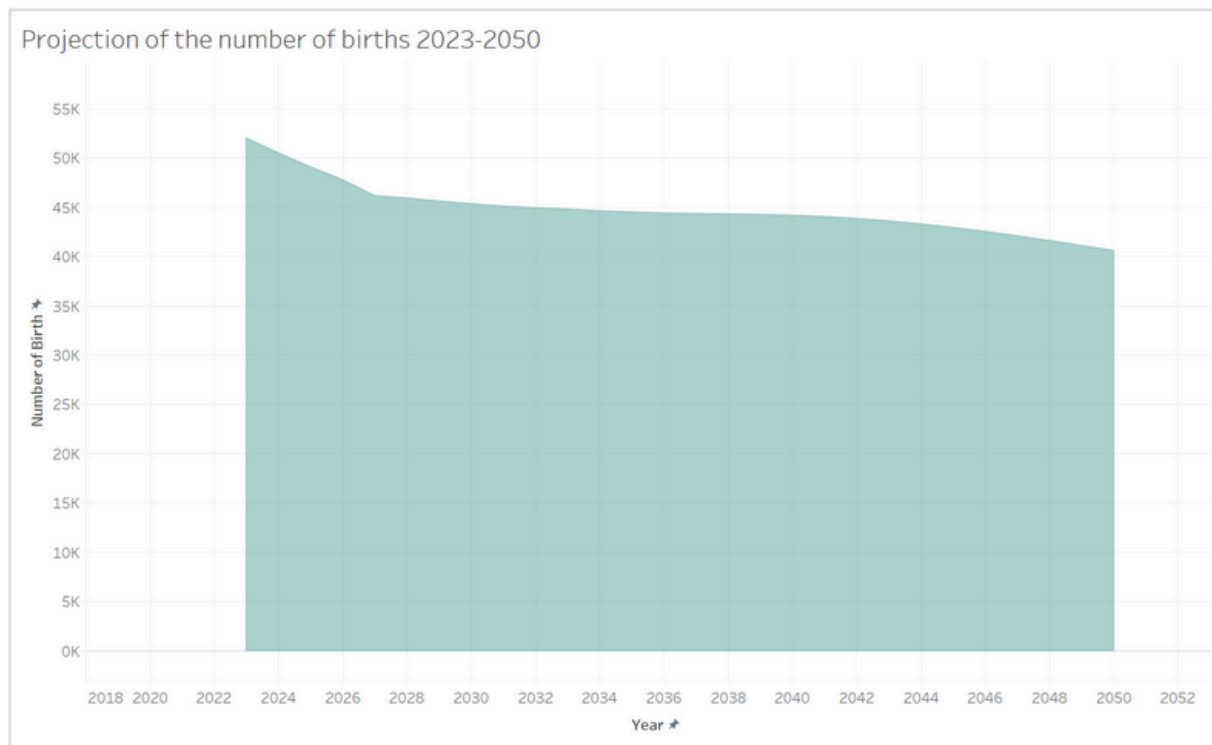


Figure 12: Projection of the number of births

In light of these demographic challenges, immigration stands out as a potential long-term solution to sustain Norway's workforce and economic stability. While immigrants currently make up less than one-fifth of the population (17.3%), their role in society is becoming increasingly important.

In the following section, we take a closer look at how immigrants actively participate in Norwegian society, through education, employment, and entrepreneurship. These insights are supported by data visualizations that reveal important trends and patterns, helping us understand their contribution and potential even better.

Education

Higher education is something many immigrants choose to invest in, ranging from elementary school all the way to university studies. This is a crucial investment that enhances their skill and improves their prospects in the difficult job market.

In addition to the academic education, immigrants participate in specialized language courses to faster and easier learn the language. This initiative is designed to bridge the cultural and communication gaps, making the integration process easier for the immigrants.

Through the education, they will acquire both practical and theoretical skills and knowledge, making it possible for them to have a strengthening impact on Norway's workforce.

And as we can see from the graph below, there is a trend upwards for both men and women that gain basic education and higher education, increasing each year.

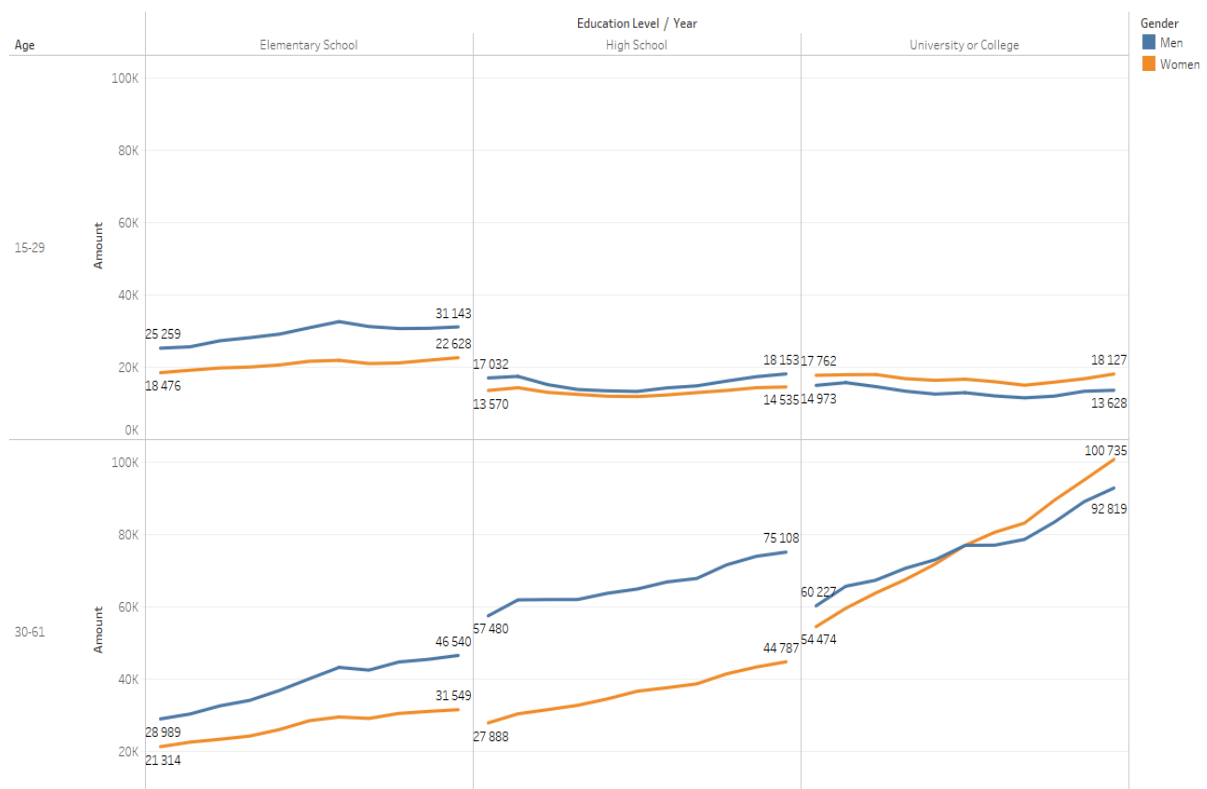


Figure 13: Education level per year, excluding 62+

To show the details for the over 62 years old category we had to exclude them from the others as their numbers are, as you can assume, significantly smaller compared to the younger people.

Education level pr. year 62+

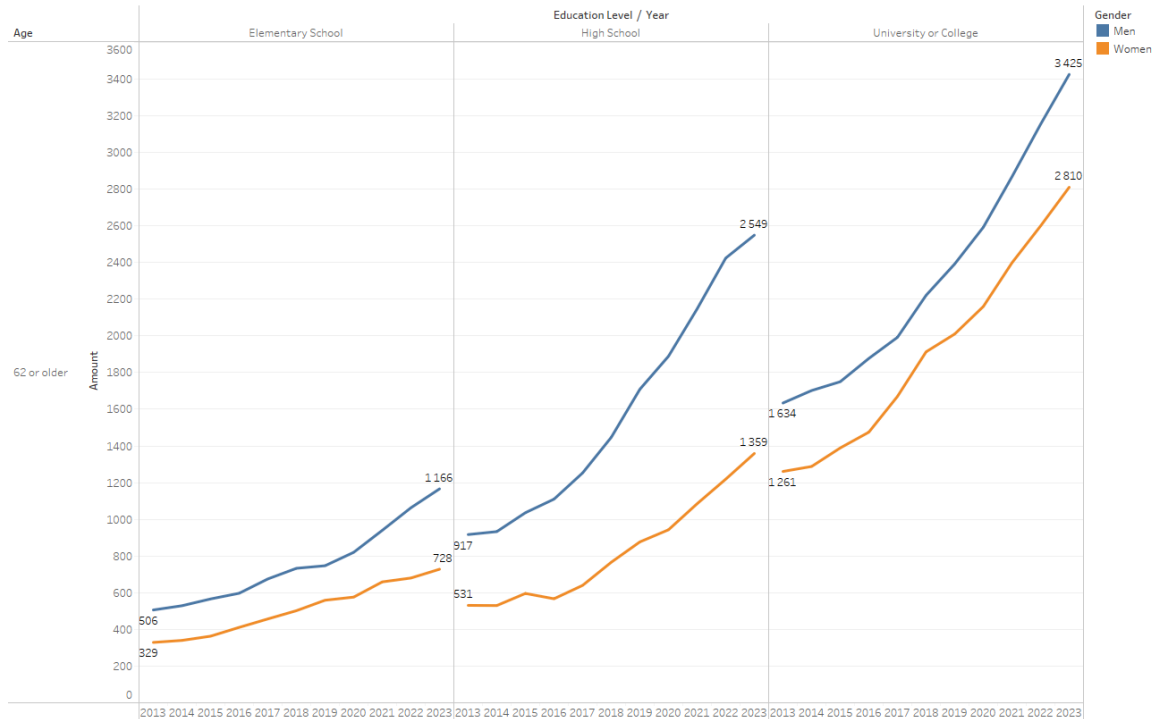


Figure 14: Education level pr year for 62+

Employment

A large number of immigrants secure employment soon after they arrive in the country or when they complete their studies, to support themselves and their family. And there is also an increasing trend upwards for both men and women. The numbers for unemployed immigrants had a spike in 2020, most likely caused by COVID, this is also reflected in the Employed Only column where we can see a clear dip in 2020. But in the following years it has been trending downwards which is a positive sign.

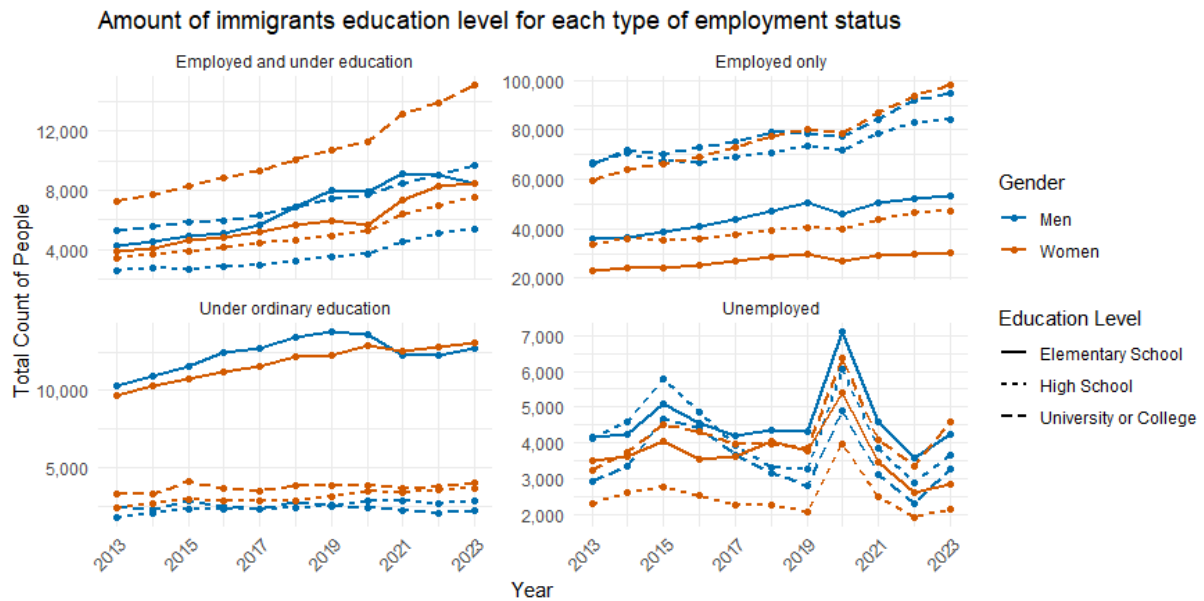


Figure 15: R visualization faceted line plot: Education level and employment status

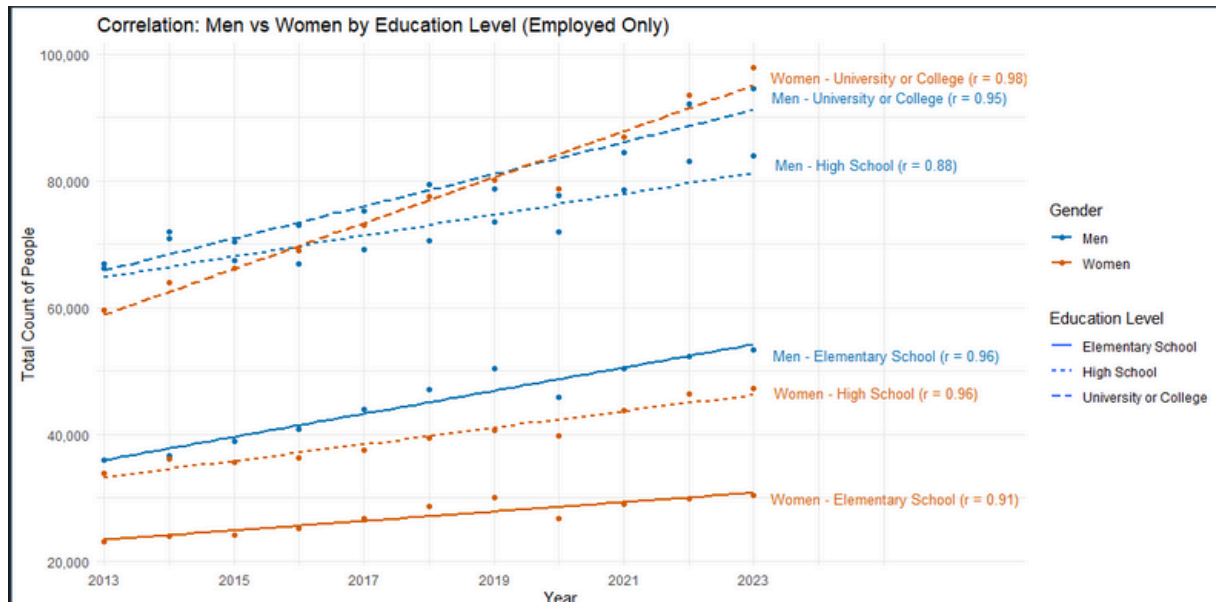


Figure 16: RStudio Correlation: Men vs Women Education Level (Employed Only)

This analysis looks at the amount of immigrants which are listed as “Employed Only” in our dataset. The results suggest and showcase a strong positive correlation across most of the employed only category, especially for the people who have taken a higher education. Let’s see especially on immigrant women with a university or college degree, who show a correlation of 0.98, which indicates a clear and steady increasing trend over time. Looking at the men's side of the university or college education we can see a similar trend of steady positive trends, with a r value of 0.95. As a result of this we can confidently assume that the higher education you have leads to having a higher chance of getting employment, and that the gender gap between men and women is closing rapidly even seeing women jump ahead for the highest educated category.

Total aggregated statistics, for education and employment:

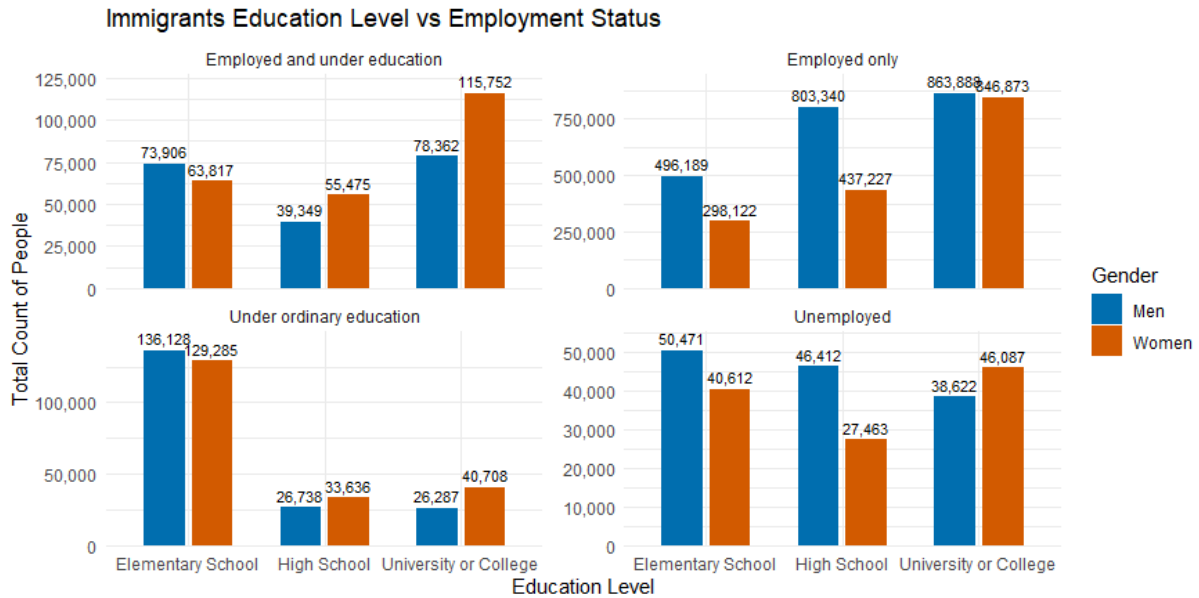


Figure 17: R visualization bar chart: Education level vs Employment Status

Entrepreneurship

A significant proportion of the immigrants take the entrepreneurial route by establishing their own companies. This creates economic growth by introducing innovative products and services, and also creates new job opportunities.

Immigrant entrepreneurs often bring fresh perspectives and expertise to the market. Their businesses lead to diversifying the marketplace and may serve unmet needs within their communities. As we see from the graph below, the majority of businesses established by immigrants are the Enkeltpersonforetak (ENK) with the majority being established by men.

Also we discussed to only focus on the organization form ENK (Enkeltpersonforetak) which is the most prominent one, the other forms of organization DA (Delt Ansvar) and ANS (Ansvarlig Selskap) had some entries but it is not noticeable really, but we thought it would be fine to add to see the difference. Also, we can see that the age group 25-44 is where the most action happens when it comes to starting businesses.

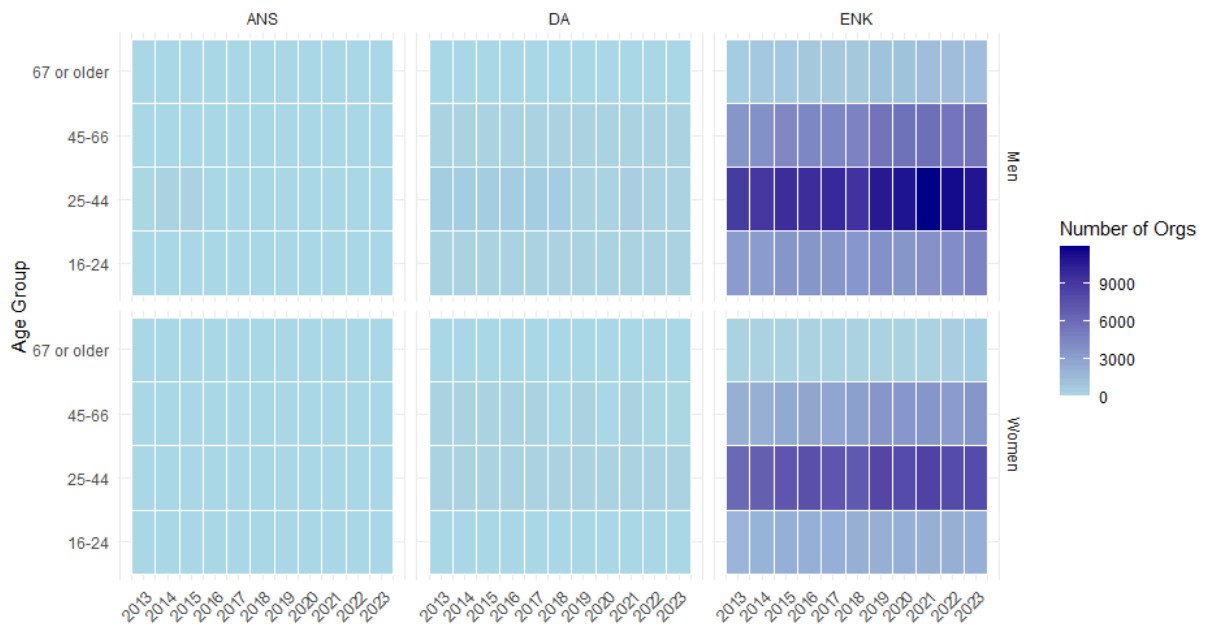


Figure 18: R visualization - Heatmap Amount of organizations started by immigrants

However, immigration is not a simple solution either.

According to projections from Statistics Norway (SSB), immigration is expected to gradually increase in the coming years. At the same time, the data show that **net immigration** (immigration minus emigration) is projected to **decrease significantly** in the coming decades. In 2024, net immigration was 41,093 people, but by 2050 it is expected to fall to around 13,000. This suggests that **emigration may also increase**, raising important questions.

The reasons behind this trend remain unclear, and no official reports have yet been published to explain the development. It is a topic that is still under research and investigation.

The graphs below illustrate the projected trends for immigration, emigration, and net immigration in the years ahead.

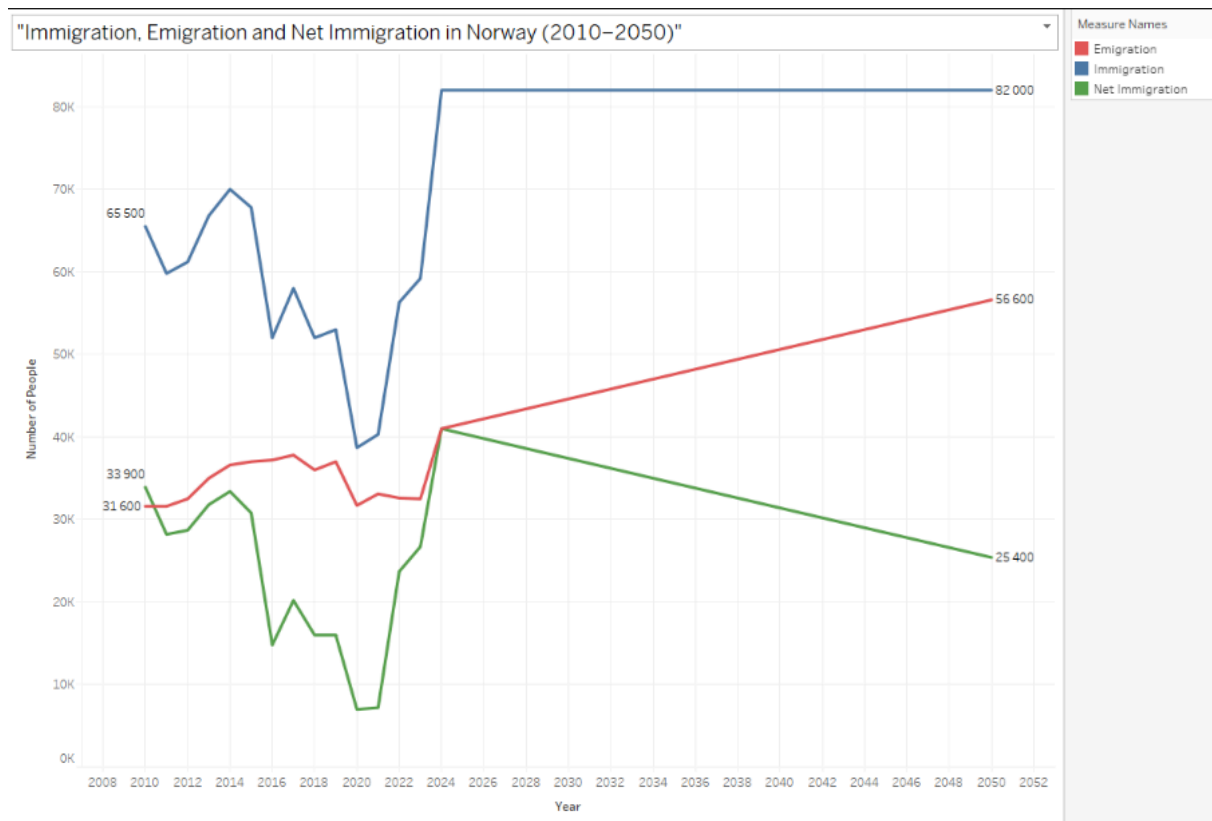


Figure 19: Projected Immigration, Emigration and Net Immigration

Due to increasing emigration and an aging population, Norway should adopt a more strategic approach to address the upcoming demographic challenges. In the following, we take a closer look at the regulations and trends related to immigration to Norway.

According to Norwegian regulations, individuals who wish to live in Norway for more than three months must register or apply for a residence permit. There are different types of permits, including family immigration, work, protection (asylum), and studies.

The Bar chart below shows the total number of immigrants by continent and type of residence permit from 2013 to 2023. Asia and Europe are the two largest sources of immigration to Norway.

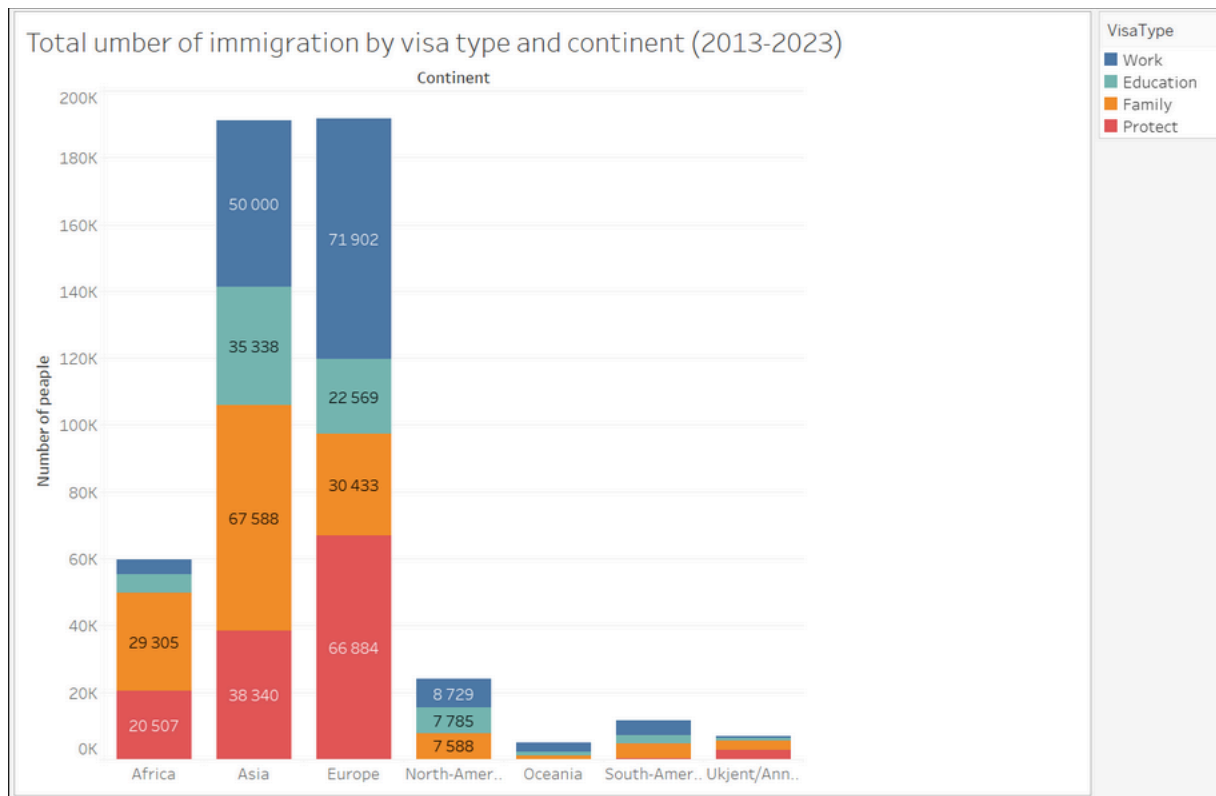


Figure 20: Total number of immigration by visa type and continent (Stacked Bar)

From Europe, work permits have been the most common type of residence permit throughout most of the period, while asylum permits increased significantly from 2021 due to the war in Ukraine. From Asia, family, work, and asylum permits have been the most common, with a clear peak in asylum immigration between 2015 and 2017 as a result of the war in Syria.

These patterns illustrate how international conflicts influence asylum immigration, while the underlying need for labor in Norway is reflected in the high number of work permits, particularly from Europe.

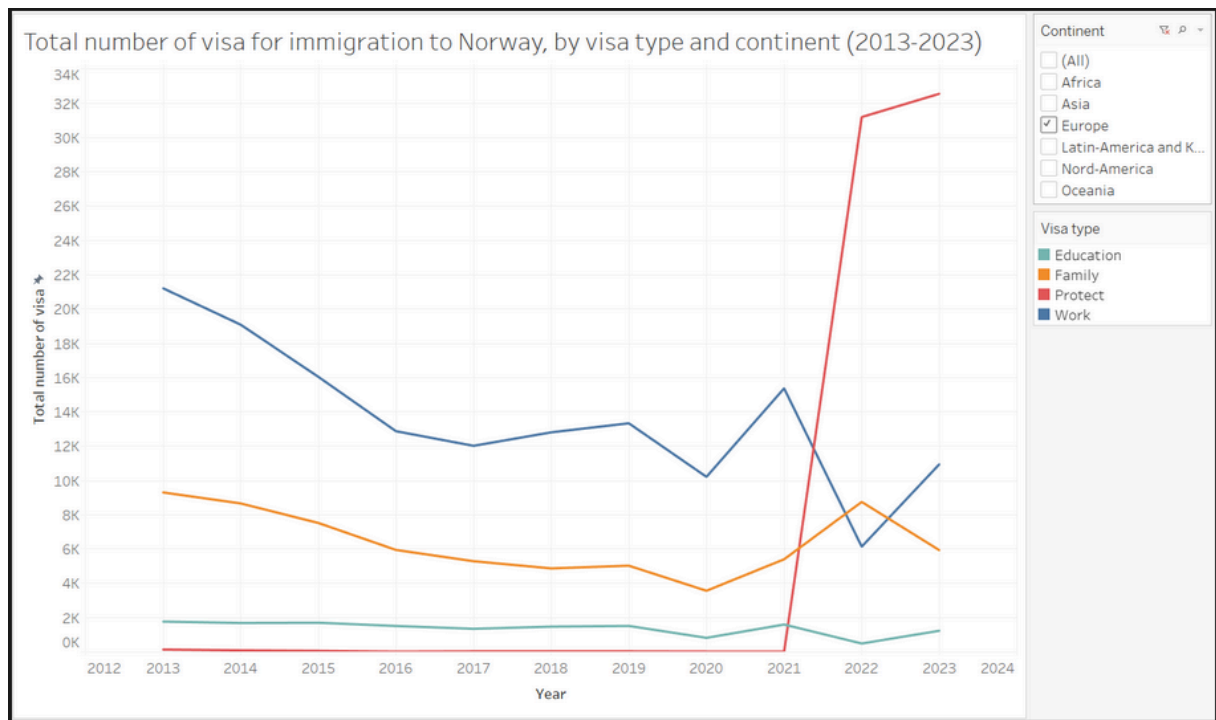


Figure 21: Total number of immigration by visa type and continent (Line Graph)

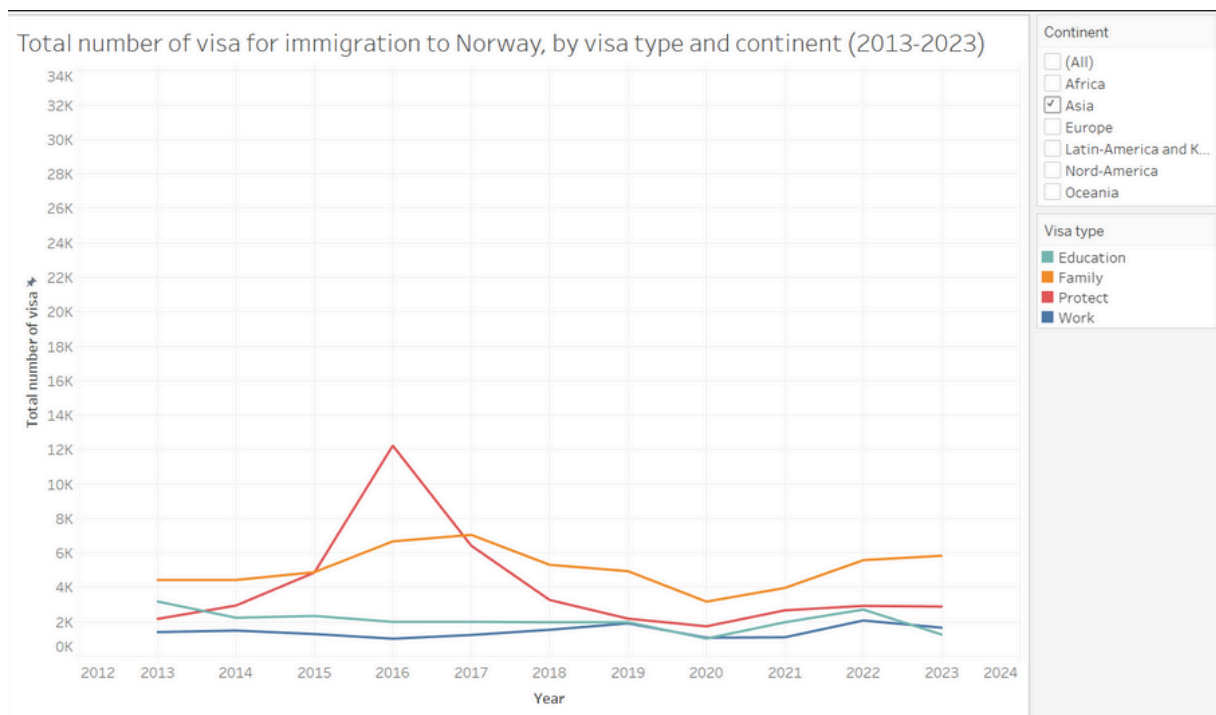


Figure 22: Total number of immigration by visa type and continent (Line Graph,

Filtered by Asia)

(Same scale used in the graph for better comparison)

The graphs also show that many students, especially from Asia and Europe, choose to study in Norway.

According to data from the Directorate for Higher Education and Skills, the number of international students increases each year, except for 2020 and 2021, when the pandemic caused a temporary decline.



Figure 23: Total number of international students

International students represent an important resource for Norwegian society. They are young, highly educated, and already living in the country. According to Norwegian regulations, they may apply for a work permit after completing their studies if they secure a job. This makes them a valuable group that can help meet the country's growing demand for labor.

However, figures from Statistics Norway (SSB) show that nearly 6 out of 10 international students from countries outside the EU and EEA, leave the country after completing their studies. Since employment is a key reason for staying in Norway, the majority of international

students who wish to remain in the country enter the workforce quickly and end up staying for several years. The table below shows that nearly 86% of third-country students who studied in Norway between 2013/2014 and 2017/2018 were employed in 2023 (Bartsch, 2024).

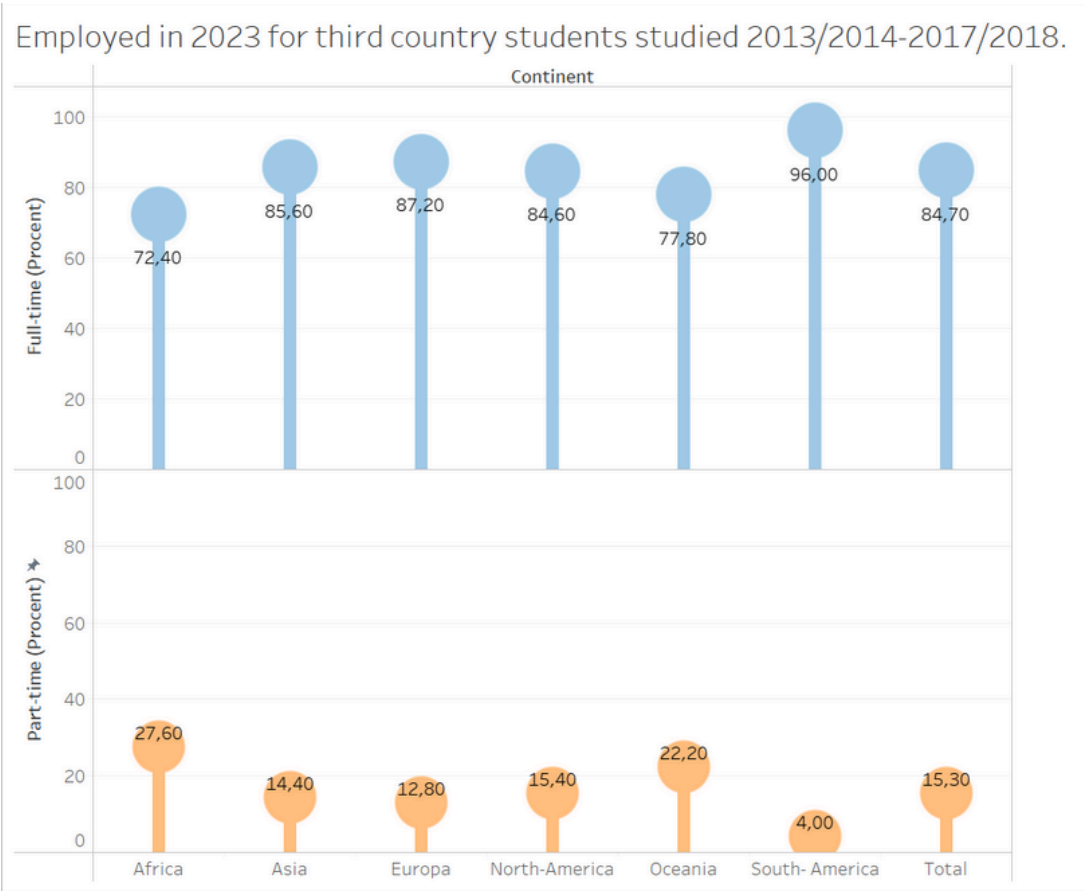


Figure 24: Employed in 2023 for third country students

If Norway succeeds in developing effective strategies to retain more of these students, it could make a significant contribution to meeting the labor market demands in the coming years.

After preparing the data and identifying the demographic challenges related to an aging population and labor shortages, it is important to consider how immigration is viewed as a potential solution in the Norwegian context. To understand this, we take a closer look at how immigration and immigrants are perceived in Norwegian society.

Immigration Thoughts

Over the course of time, Norwegians' perspectives on immigration have shifted, notably with regard to refugees' policies, employment opportunities, cultural norms, and safety concerns.

Many Norwegians believe that immigrants have a positive impact on their work lives. In the year 2023, the percentage of people who concurred with this statement increased from 72% in the year 2015 to 87%. Although it fell to 82% in 2024, the majority of people continue to believe that immigrants are beneficial to the job market. In 2024, a mere six percent of the polled individuals expressed that immigration had no positive effects.

Most immigrants contribute to Norway's vibrant cultural life. The percentage of Norwegians who agreed with these findings jumped from 71% in 2015 to 79% in 2023. Despite falling slightly to 75% in 2024, the overall trend remains upward. There is a drop in the number of those who disagree. 14% of those who responded in 2015 disagreed that immigrants contributed to their culture, but by 2023, that number had fallen to 8%. The percentage increased a little to 11% in 2024, although generally speaking, more individuals now acknowledge the cultural advantages of immigration.



Figure 25: Norwegian View on Immigration

The general public's perception of whether immigration has an impact on safety has evolved throughout time. 26% of respondents said in 2015 that immigrants made Norway less secure. The rate fell to 14 percent by 2023. It did, however, rise to 22% in 2024, indicating that some worries have reappeared. A majority of Norwegians, however, do not perceive immigration as a threat. Seventy-two percent of respondents in 2023 disagreed that immigration reduces Norway's safety. Even though the percentage fell to 66% by 2024, most people still believe that immigration poses no threat to national security.

Attitudes about asylum and refugee policies have evolved. The number of Norwegians who support facilitating the stay of refugees has increased. The percentage of people who wished to loosen limitations rose from 15% in 2015 to 22% in 2023. It decreased somewhat to 18% in 2024. Support has been erratic for tougher refugee laws. In 2015, 29% of respondents desired stricter asylum laws; by 2023, that number had dropped to 9%. In 2024, it increased to 19%, indicating a resurgence of support for stricter laws



Figure 26: Refugee Policy

Most people still want things to remain the same despite this. Half of the respondents thought that the refugee policy should not change in 2015. In 2024, that percentage was still comparatively high at 51%, having increased to 58% by 2023.

7 Conclusion

This project was to investigate how immigration might help Norway deal with its demographic issues, specifically the widening gap between an aging population and a declining labor force. We have demonstrated through data analysis and visualizations that immigration plays an important role in both population growth and the long-term viability of Norway’s welfare and economic systems.

We discovered a number of significant trends. The first trend was that while the number of young people in Norway is expected to stay the same or even decline, the number of elderly people is expected to rise dramatically over the next several years up until 2050. This demographic mismatch puts more strain on the social and healthcare sectors and raises the possibility of a labor shortage. Immigration comes in as a necessary solution to fill the gaps, with the work-related immigrants providing immediate contributions to the labor force while immigrants coming for education provides a long-term potential, if proper retention strategies are improved upon.

We also found that Norway's annual immigration is likely to remain stable, the net immigration is expected to decline caused by the rising trend in emigration. This shows the need for policies that not only attract but also retain immigrants by making improvements in integration policy and in expanding employment opportunities. As showcased in the visualization and analysis of different visa types it was clear that work visas provide the immigrants with the most direct benefits, but family and education visas need more assistance to have the same benefits.

The public have a generally good view on immigration and while there have been some issues and concerns around safety and refugee policies in recent years, the public are more positive towards immigrants now than in previous years. This suggests that while immigration is increasingly accepted, political and social discussions will still continue to affect how immigration is managed.

While we look ahead, Norway's immigration policies should focus on three main pillars:

- (1) enhancing the integration of immigrants into the workforce, particularly women and family migrants,
- (2) creating stronger incentives to retain international students and skilled workers,
- (3) preparing for a global migration shift driven by factors such as climate change and geopolitical instability.

In conclusion, immigration is and will remain a part in influencing Norway's future. It is not only about the amount of immigrants, but about developing even more inclusive policies that ensure both immigrants and the society at large can develop together. To handle the

opportunities but also the challenges that come with immigration, it will require a sustained investment in proactive policy making based on data.

8 Dashboard

Description and purpose

Filename: Visualisering av data Dashboard.twbx

This Dashboard is made using Tableau, and it presents central society data related to immigration and integration in Norway by the period of 2014 - 2023. The visualizations are made using datasets from SSB and UDI, and they cover three main topics:

Immigration to Norway by Visa Type: Map Display over which county immigrants come from, distributed by Visa Type.

Employment Status by Age Group and Gender: Line Chart which shows the development in employment status over time, categorized by gender and age group.

Organizations Established by 25-44 year olds by Education Level: Line Chart which shows the organizations that are started by individuals between the year 25-44 year olds based on their education level.

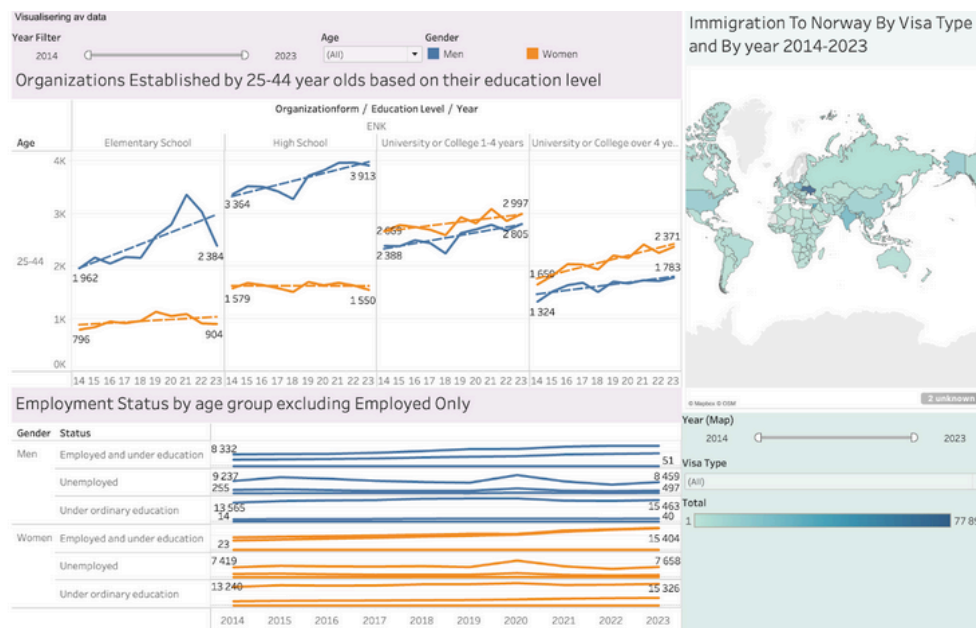


Figure 27: Dashboard

The purpose of the dashboard is to give a comprehensive and visual understanding of immigration demographics and how different groups contribute in the society by work and entrepreneurship. The

dashboard is interactive and gives the user the ability to filter by year, gender and age.

Usability

The Dashboard is structured in a clean and logical way, with clear division of information in three main visualizations. The user gets a clean overview without the need to interpret graphs with the right colors and text size. The map to the right gives a geographical entrance to the data, while the line graphs to the left shows the development over a specific time frame in an easy and understanding way.

The interactive filters for year, gender and age are placed on the top of the page that interacts with the line graphs and we chose to show that by color pink, while the map has its own filters marked in blue. The use of colors and axis is consistent and filters allow the user to filter years across the line graphs even if they come from a different dataset.

For example, the user can easily see how the unemployment between females in the age of 25-44 have developed over time and compare it to the education level and establishment of new organizations. The dashboard is very useful for decision makers in the government and the education sector, because it gives a fast understanding of contributions and demographic groups that contribute to Norwegian society.

9 Project management

To succeed with this project, it was important to divide the work so that everyone could contribute. The project was organized based on the research questions, and each team member was given individual responsibility for preparing the data and creating the visualizations related to their assigned part of the project. This allowed everyone to work with data cleaning, visualization, and writing.

We further divided the work so that some team members focused mainly on working in Tableau, while one person worked in R to meet all the project requirements. This division of tasks worked well, but we supported each other whenever different challenges arose along the

way. The project leader was responsible for monitoring progress, ensuring that everyone was engaged, and guiding the team in the right direction.

Even though the structure might seem hierarchical, the way we worked was much more open and collaborative. Everyone contributed ideas and suggestions, and we discussed different solutions together, but the project leader had the final say when decisions had to be made.

The biggest challenge we faced in this project was related to data collection and dataset management.

Since we chose to work with immigration, which is a large and complex topic, we quickly realized that there was no single, comprehensive dataset that covered all the aspects we wanted to explore. Instead, we had to rely on many smaller datasets and tables, mostly from Statistics Norway (SSB) and other sources.

Finding datasets that were relevant, reliable, and connected to our problem statement was time-consuming and often frustrating. Sometimes we found data that seemed promising, but after working with it for a while, we discovered that it didn't really fit our project or couldn't be used in combination with the other data we had. This led to several rounds of adjustments in our problem statement, project structure, and data selection. This process took a lot of time and made the project more complex than we had originally expected.

Another major consequence of using multiple, unrelated datasets was that it became very difficult to build a fully **interactive dashboard**. Since most of the datasets were not directly linked or structured in the same way, it was not possible to create interactivity between all the different visualizations. This limited the user experience and made it challenging to present the data in a way that felt fully connected.

If we were to do this project again, we would definitely consider choosing **a single, larger dataset with multiple variables**, even if that meant narrowing the scope of the project. This would have made the data management easier, saved us a lot of time, and allowed us to create a more seamless and interactive dashboard.

To cover which project member did what, we chose to create a table and insert all of the members contributions and an accumulated time spent, across all tasks.

*

Figure 28: Project team structure

Group Member Contributions:

Member	Description	Time spent
7001	Manage the project and assign responsibilities Developing the project structure and writing the project proposal Defining the topic and identifying relevant datasets Writing the problem statement, research questions, the rational and target audience Planning the data storytelling and report outline Creating the visualizations shown in Figures 7, 8, 11, 12, 20, 21, 22, 23 and 24	60 hours
*	Background of topic Infographic Methodology and Tools Aging Population and need for immigration in norway, Employment need Formatting the report	55 hours
*	Dataset cleaning and preparation Description of dataset Immigration Thoughts Creating Visualizations Dashboard	40 hours
*	Immigration and population data Rationale and Target Audience of the problem statement Data preparation Emigration prediction and net immigration calculation Creating the visualizations shown in Figures 4 and 5 Clear accessible writing style	42 hours
*	R Studio Design choices, principles Education, Employment, Entrepreneurship Project Management Conclusion Final edits of report	45 hours

*

I was responsible for writing the background of the topic and explained why immigration is important for Norway's future. I also wrote the methodology section, in which I described how we used Excel for initial review of the datasets, Tableau Prep for cleaning and merging the data, and Tableau to build visualizations and dashboards. In addition, I conducted an analysis on how Norway's aging population affects the need for immigration. This involved applying regression analysis in Tableau to forecast demographic trends. I also created an

infographic that summarizes our key findings in a clear and visual format to make the insights more accessible to our target audience. Lastly, I was in charge of formatting the report, including the table of contents and ensuring overall consistency in structure and design throughout the document.

*

I got to work with our data to get a good overview and understand what we were working with. Then, I started creating visualizations quickly because I wanted to start with the theory and analysis as early as possible. The goal was to use the visualizations to see patterns and trends in the data, which we could later analyze in more depth. I have always focused on making the visualization easy to understand, both for ourselves and for others who might be looking at our results. It is about making the data more accessible and clearer so that it is easier to conclude further down the process.

7001

As team leader, I contributed to defining the project's topic and identifying relevant and coherent datasets. The group had a general idea of working on the topic of immigration, but it was my responsibility to clarify the problem statement and define the scope of the project to ensure that it would be both relevant and coherent. This turned out to be a challenging task.

Immigration is a large and broad topic, and we did not find one large, comprehensive dataset that covered all the aspects we wanted to explore. Instead, the available data consisted of many small datasets and tables. Finding datasets that supported the different parts we wanted to include in the report was therefore not easy. Often, we managed to develop a good problem statement and a solid structure for the storytelling, but we lacked data that actually supported it. Other times, I found datasets, but they were not connected or relevant to each other, making it difficult to create a coherent and meaningful story.

This led to several adjustments in the problem statement and the project plan along the way. Although the project sometimes felt a bit unclear in the beginning, I eventually managed to define a problem statement and find datasets that supported each other and allowed us to build a coherent narrative.

In addition to this work, I spent a lot of time processing and preparing data in Tableau Prep and creating visualizations in Tableau Desktop. This provided me with valuable experience using both tools.

*

Starting this project felt a bit daunting, since we had to come up with an idea and gather all of the data required. Being creative and imaginative felt a bit restricting in this part of the project, as a lot of ideas were flying around without us landing on a specific idea.

My work within the project was related to the design principles and choices, which set the expectations for what the reader of the report is gonna be seeing in the visualizations. Following this I was responsible for creating visualizations for the topics Education, Employment and Entrepreneurship. At the start these were done in Tableau, but we decided to pivot off this idea and use R for this part, since we didn't have any R related things featured as of yet. I personally found R to be more flexible than Tableau, allowing me to make adjustments and tweaks to the code and see instant results of the tweaks. Also, using R I assisted the group with creating the correlation analysis so that we could include them in the report.

Overall, this project was a valuable experience in teamwork and expanding on my technical and analytical skills. Although it was a challenging project, I think we were able to succeed in creating a good end product.

*

I worked mostly with immigration and population data, and looked at how these numbers are expected to change in the years ahead. I worked on the rationale and target audience of the problem statement and one of the research questions, and tried to explain why this topic matters for Norway. I also helped clean and prepare the data in Excel and Tableau Prep so we could use it in our visualizations.

Since there were no official numbers for how emigration might change in the future, I made my own prediction based on older data. I then used that together with numbers from SSB to find out how net immigration might look in the future. I created two visualizations in Tableau to show this and wrote the text that explains what the data tells us. I also worked on making the language in our text simple and clear, so that people outside of school or research could also understand what we found.

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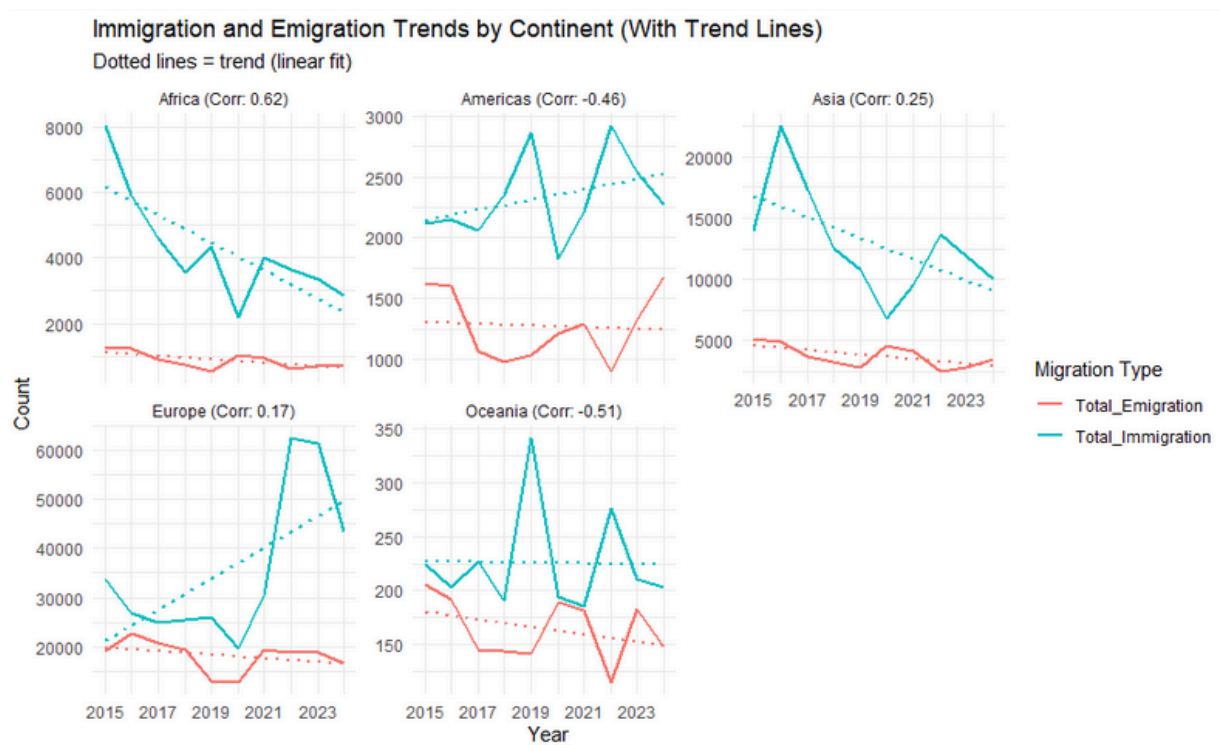
11 Appendices

- **Infographic:**

<https://www.figma.com/design/BNibzslglnI56cODfB0b2O/Infografikk?node-id=33-2&t=49epFilOT1NPRmWa-1>

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Unused Visualizations:



R Correlation